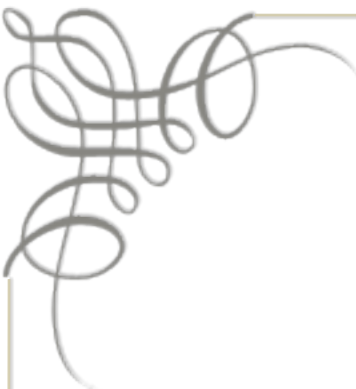




Projective Geometry and Transformation of 2D

CPS 843 (CP 8307)

Introduction to Computer Vision

Richard Wang



Logistics

Upcoming Deadlines ▾

Richard Wang posted on Sep 7, 2025 9:43 AM

-  Edited

- **Quizzes:** Quiz#3 (10/19-20)
 - **Assignment:** HW#2 (10/20), HW#3 (11/03)
 - **Tutorials:** Tutorial wk6: TA office hour (10/10)
 - Link: <https://torontomu.zoom.us/j/94831563369>
 - **Group enrollment:** (11/03)
 - **Project presentation/progress report:** (11/24)
 - **Project report:** (12/08)
 - **Final exam:** tbd
-
- HW-2: Monday 10/20. Quiz#3: 10/19-20
 - Group enrollment: 11/03 (both CPS843 and CP8307)

Least Squares Data Fitting

Least Squares Line Fitting

- Forms of equations of a straight line

- General (standard) form

$$ax + by + c = 0$$

- Slope-intercept form

$$y = kx + b$$

- Point-slope form

$$y - y_1 = k(x - x_1)$$

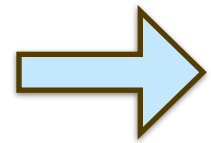
- Intercept form

$$\frac{x}{a} + \frac{y}{b} = 1$$

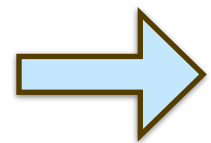
Least Squares Line Fitting

- Find a line passing through two points (x_1, y_1) (x_2, y_2)

$$y = kx + b$$

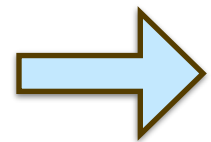


$$\begin{cases} y_1 = kx_1 + b \\ y_2 = kx_2 + b \end{cases}$$



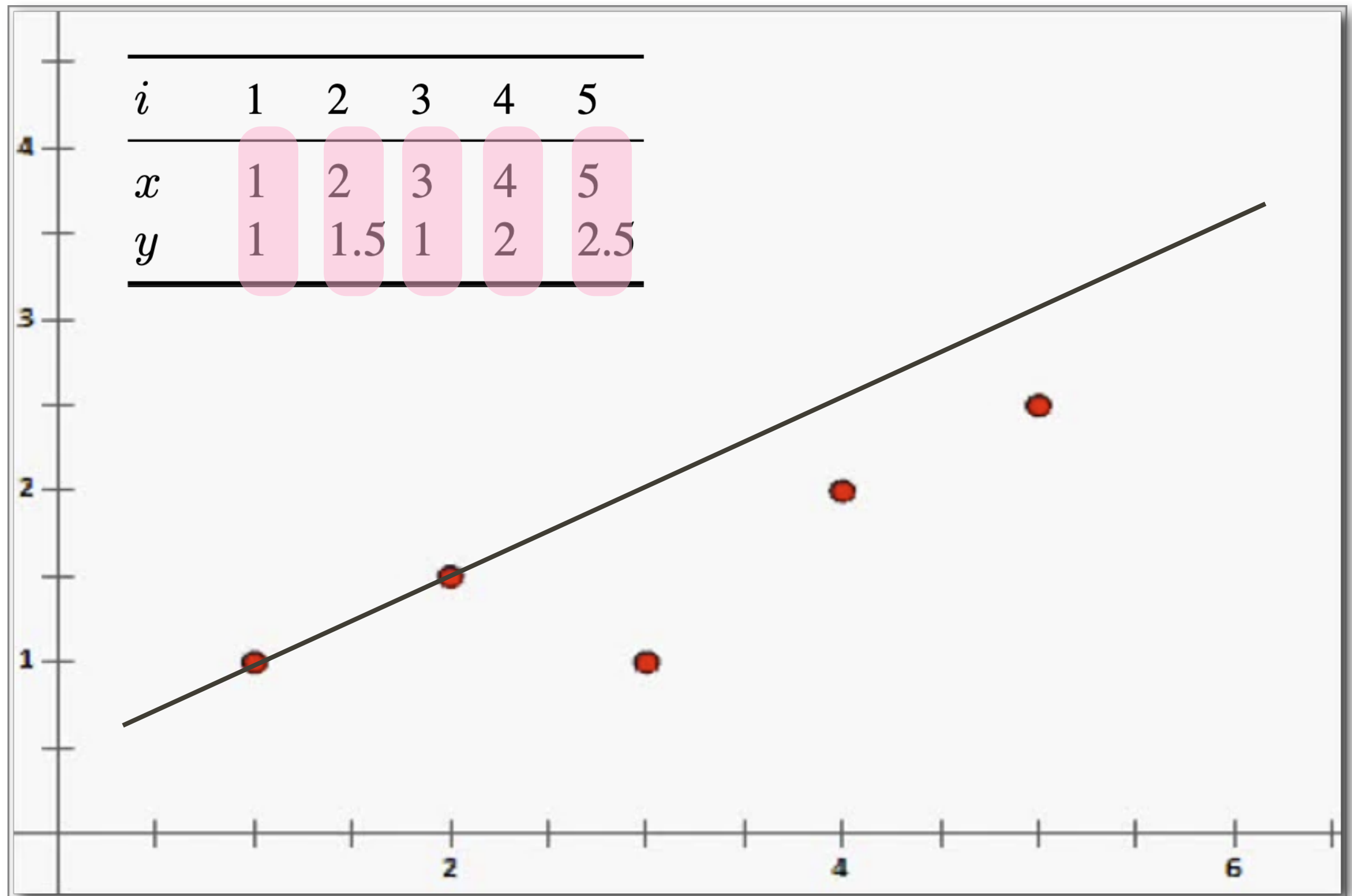
$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}_{\mathbf{y}}$$

$$\mathbf{A}\mathbf{c} = \mathbf{y}$$

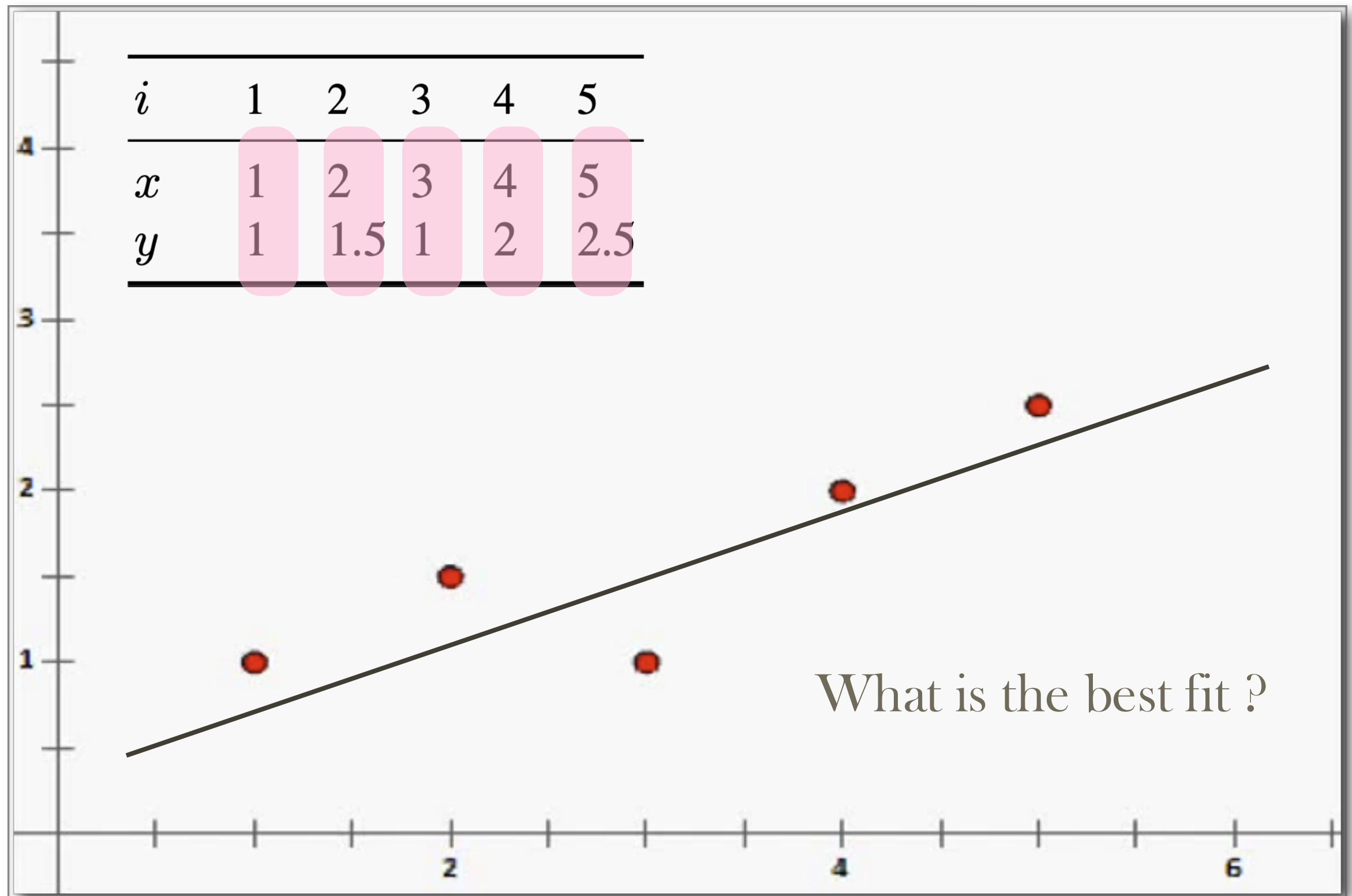


$$\mathbf{c} = \mathbf{A}^{-1}\mathbf{y}$$

Least Squares Line Fitting



Least Squares Line Fitting

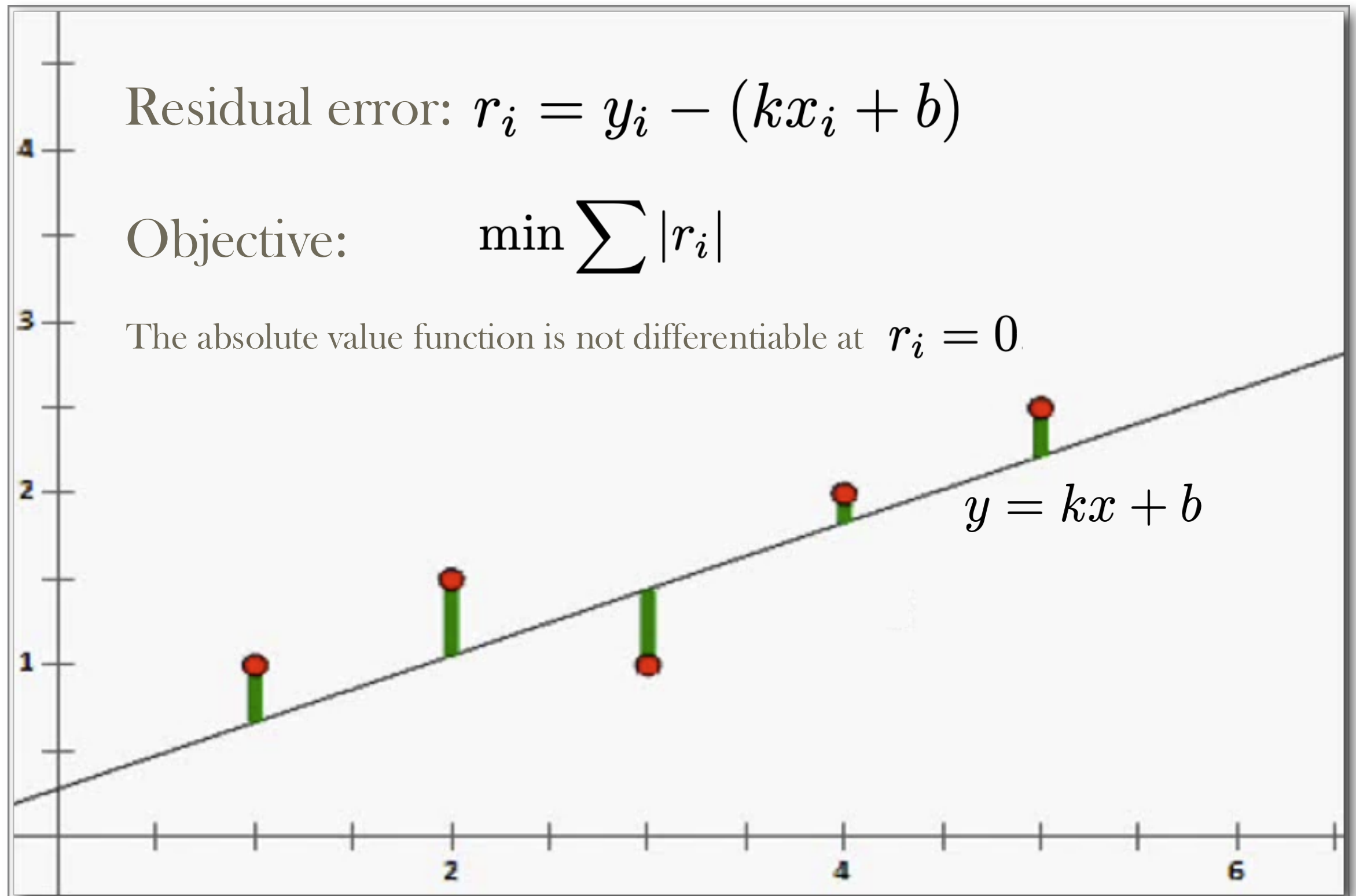


Least Squares Line Fitting

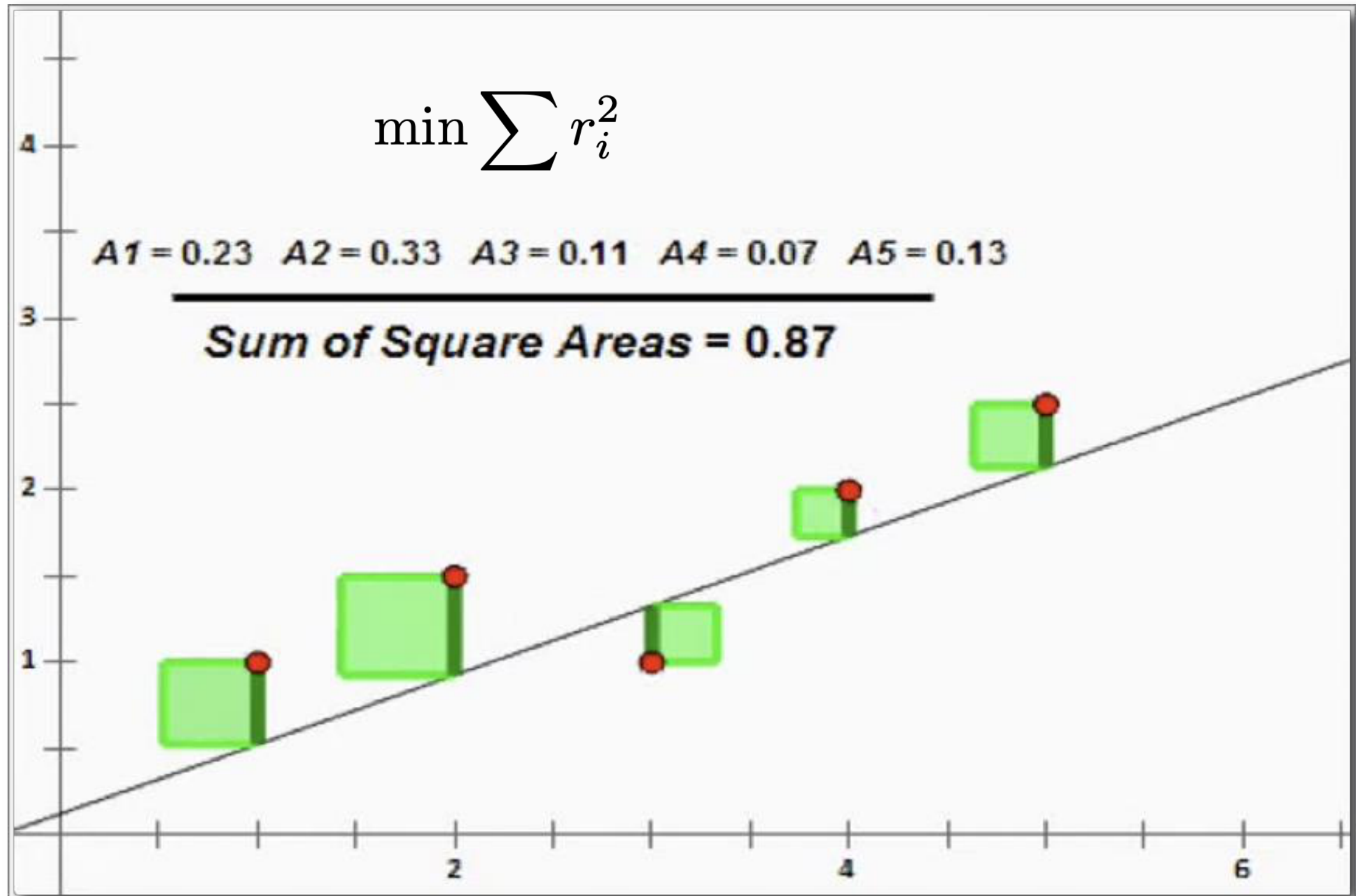
Residual error: $r_i = y_i - (kx_i + b)$

Objective: $\min \sum |r_i|$

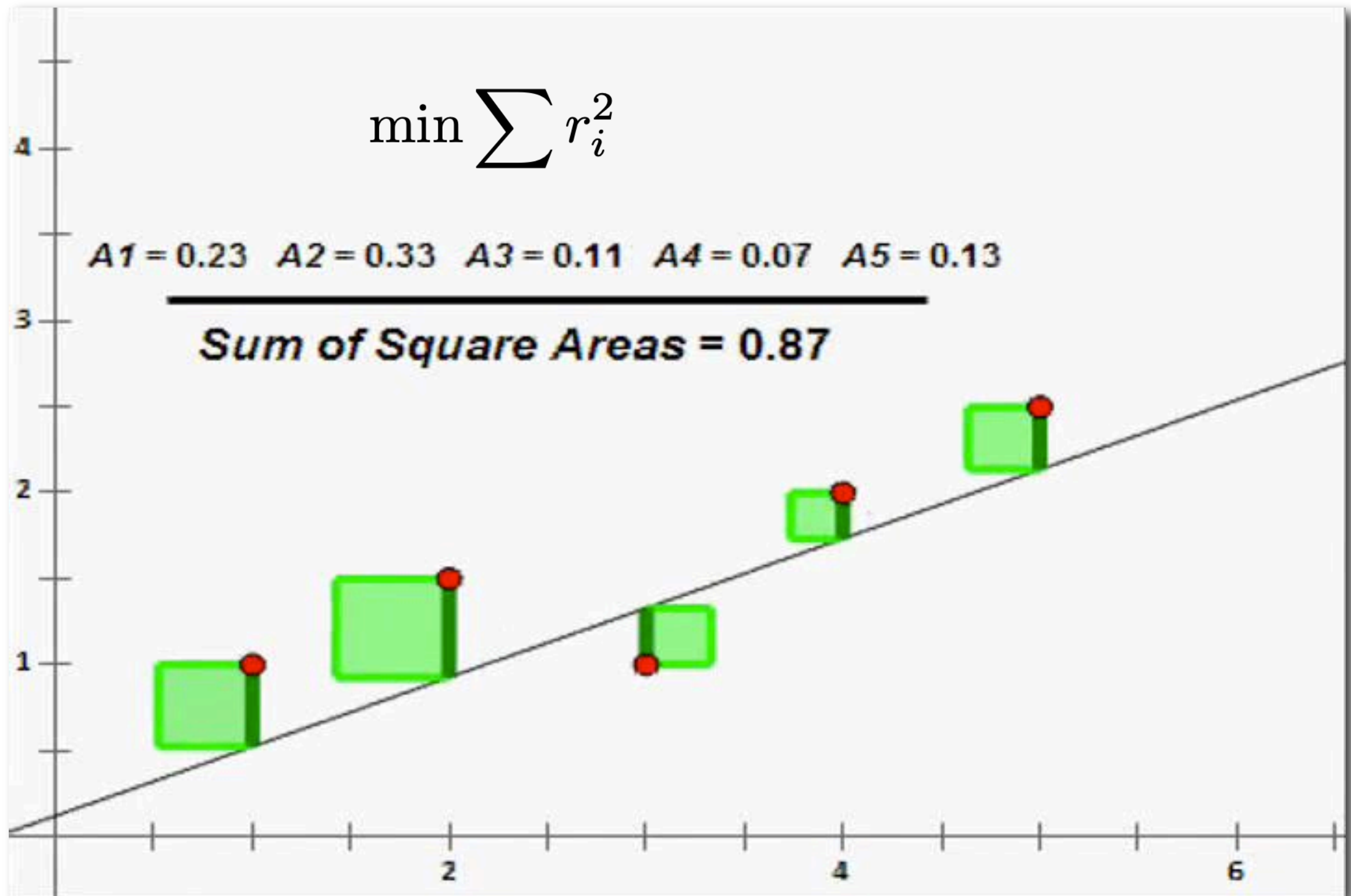
The absolute value function is not differentiable at $r_i = 0$.



Least Squares Line Fitting



Least Squares Line Fitting



Least Squares Line Fitting

- Least squares solution:

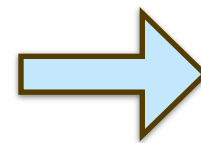
The overall solution minimizes the sum of the squares of the errors resulted from every single point.

$$\min \sum r_i^2$$

Least Squares Line Fitting

- Formulation of least squares solution

i	1	2	3	$\cdots$	m
x	x_1	x_2	x_3	$\cdots$	x_m
y	y_1	y_2	y_3	$\cdots$	y_m



$$\begin{cases} y_1 = kx_1 + b \\ y_2 = kx_2 + b \\ \vdots \\ y_m = kx_m + b \end{cases}$$

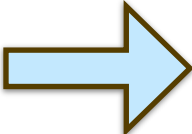
$$\begin{array}{c} \text{blue arrow} \end{array} \underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}}$$

$$\mathbf{Ac} = \mathbf{y}$$

Least Squares Line Fitting

- Formulation of least squares solution

$$\begin{cases} y_1 = kx_1 + b \\ y_2 = kx_2 + b \\ \vdots \\ y_m = kx_m + b \end{cases} \quad \Rightarrow \quad \underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}} \quad \mathbf{Ac} = \mathbf{y}$$


 Error vector: $\mathbf{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} = \begin{bmatrix} y_1 - (kx_1 + b) \\ y_2 - (kx_2 + b) \\ \vdots \\ y_m - (kx_m + b) \end{bmatrix} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}} - \underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}}$

$$\mathbf{r} = \mathbf{y} - \mathbf{Ac}$$

Least Squares Line Fitting

- Formulation of least squares solution

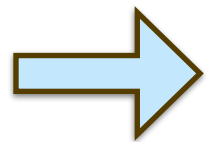
$$J = \sum_{i=1}^m (r_i)^2 = \|\mathbf{r}\|^2 = \mathbf{r}^T \mathbf{r}$$

Cost function:

$$\begin{aligned} &= (\mathbf{y} - \mathbf{A}\mathbf{c})^T (\mathbf{y} - \mathbf{A}\mathbf{c}) \\ &= \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{A}\mathbf{c} + \mathbf{c}^T \mathbf{A}^T \mathbf{A}\mathbf{c} \end{aligned}$$

Take the partial derivative w.r.t the unknown vector

$$\frac{\partial J}{\partial \mathbf{c}} = -2\mathbf{A}^T \mathbf{y} + 2\mathbf{A}^T \mathbf{A}\mathbf{c} = \mathbf{0}$$



$$(\mathbf{A}^T \mathbf{A})\mathbf{c} = \mathbf{A}^T \mathbf{y}$$

Least Squares Line Fitting

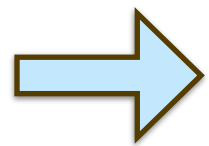
- Formulation of least squares solution (matrix form)

Normal equations

$$\underbrace{(\mathbf{A}^T \mathbf{A})}_{2 \times 2 \text{ matrix}} \mathbf{c} = \underbrace{\mathbf{A}^T \mathbf{y}}_{2\text{-vector}}$$

$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}}$$

$$\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & m \end{bmatrix} \begin{bmatrix} k \\ b \end{bmatrix} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}$$



$$\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$

$$y = c_1 x + c_2$$

Least Squares Line Fitting

- Ex.1. Fit the following 4 points into an LS line

- System equations

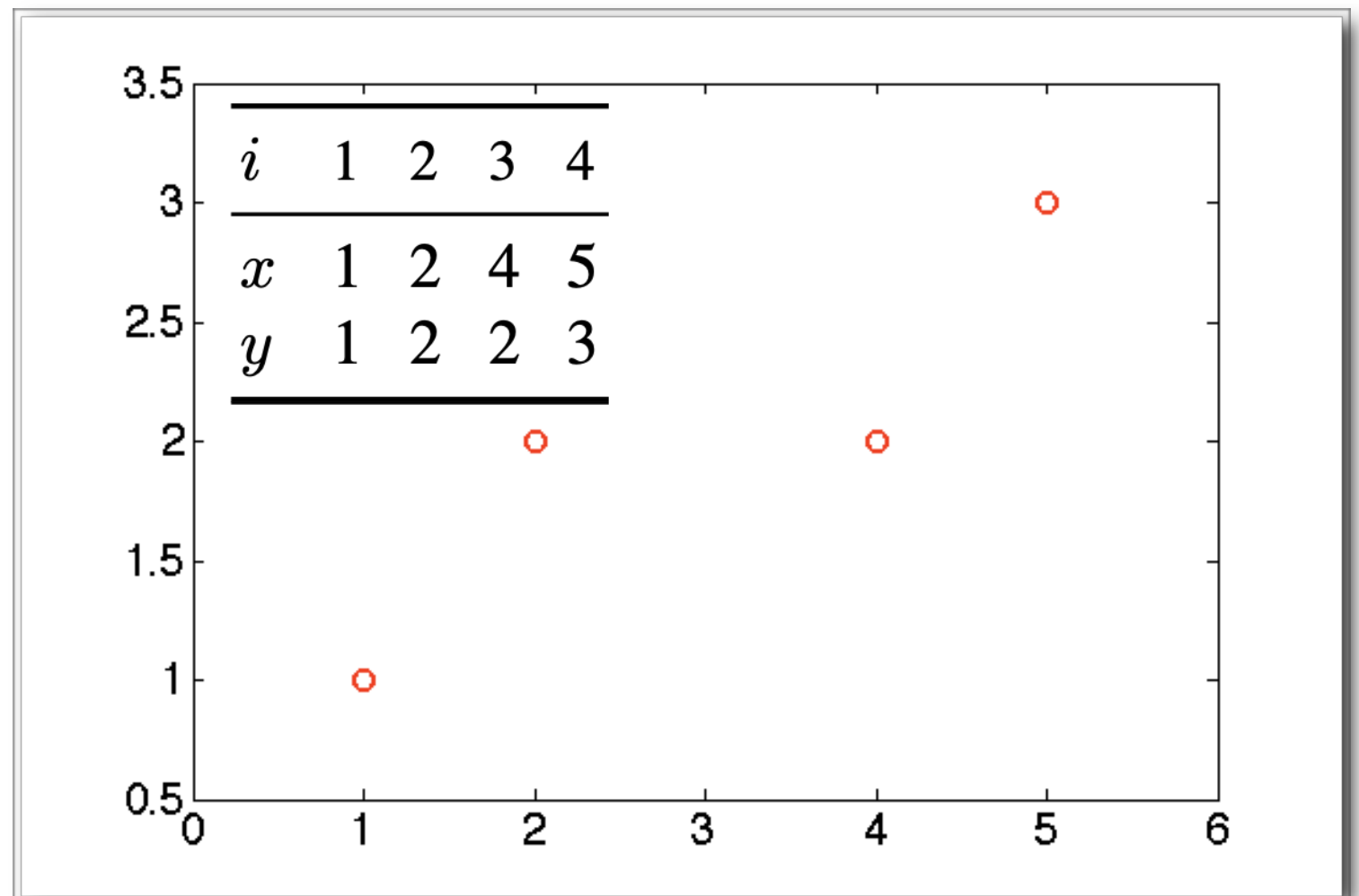
$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}}$$

- Normal equations

$$(\mathbf{A}^T \mathbf{A}) \mathbf{c} = \mathbf{A}^T \mathbf{y}$$

- Solution

$$\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$



Least Squares Line Fitting

- Ex.1. Fit the following 4 points into an LS line

- System equations

$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}}$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 1 \\ 5 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}}_{\mathbf{y}}$$

- Normal equations

$$(\mathbf{A}^T \mathbf{A}) \mathbf{c} = \mathbf{A}^T \mathbf{y}$$

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 46 & 12 \\ 12 & 4 \end{bmatrix} \quad \mathbf{A}^T \mathbf{y} = \begin{bmatrix} 28 \\ 8 \end{bmatrix}$$

- Solution

$$\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y} = \begin{bmatrix} 0.4 \\ 0.8 \end{bmatrix} \quad y = 0.4x + 0.8$$

Least Squares Line Fitting

- Ex.1. Fit the following 4 points into an LS line

- System equations

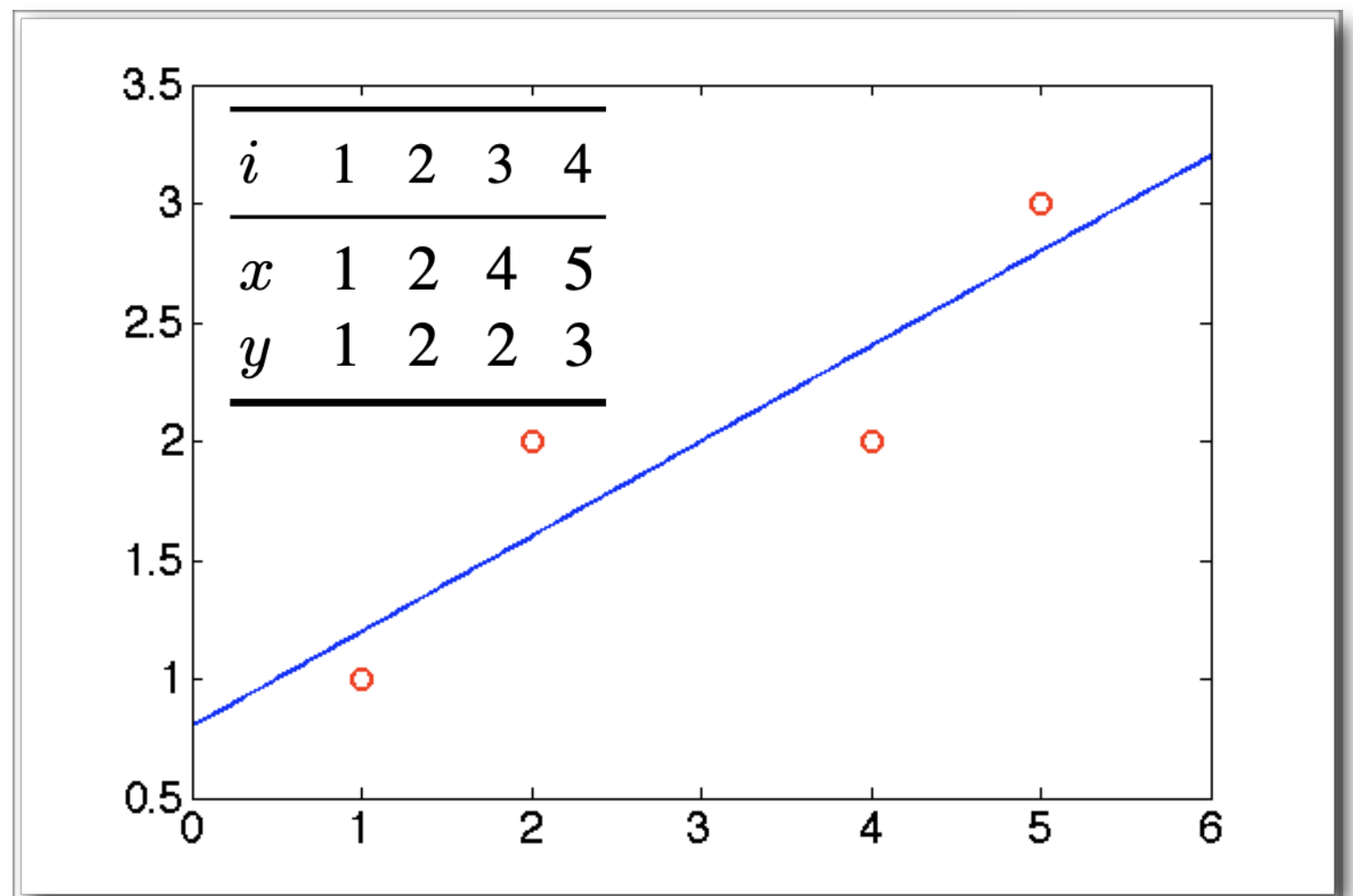
$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} k \\ b \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}}$$

- Normal equations

$$(\mathbf{A}^T \mathbf{A}) \mathbf{c} = \mathbf{A}^T \mathbf{y}$$

- Solution

$$\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$



$$y = 0.4x + 0.8$$

Least Squares Polynomial Fitting

- Polynomial equation

$$y = c_0 + c_1x + c_2x^2 + \cdots + c_{k-1}x^{k-1} + c_kx^k$$

Degree-1 polynomials (linear line equation)

$$y = c_0 + c_1x$$

Degree-2 polynomials (quadratic parabola equation)

$$y = c_0 + c_1x + c_2x^2$$

Least Squares Polynomial Fitting

- Polynomial equation

$$y = c_0 + c_1x + c_2x^2 + \cdots + c_{k-1}x^{k-1} + c_kx^k$$

Given m points, the system equations can be written as

$$\underbrace{\begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^k \\ 1 & x_2 & x_2^2 & \cdots & x_2^k \\ 1 & x_3 & x_3^2 & \cdots & x_3^k \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^k \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_k \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{y}} \quad \mathbf{Ac} = \mathbf{y}$$

Least Squares Polynomial Fitting

- Formulation of least squares solution

System equations $\mathbf{A}\mathbf{c} = \mathbf{y}$

Normal equations $(\mathbf{A}^T \mathbf{A})\mathbf{c} = \mathbf{A}^T \mathbf{y}$

LS solution $\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$

	line	parabola	cubic curve	...	degree-k
$\mathbf{A}$	$m \times 2$	$m \times 3$	$m \times 4$	...	$m \times k$
$\mathbf{A}^T \mathbf{A}$	2×2	3×3	4×4	...	$k \times k$
$\mathbf{c}$	2	3	4	...	k

Summary

- Computational details

$$\mathbf{A}\mathbf{c} = \mathbf{y}$$

- Formulate system equations

$$(\mathbf{A}^T \mathbf{A})\mathbf{c} = \mathbf{A}^T \mathbf{y}$$

- Compute the normal equations

$$\mathbf{c} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$$

- Compute LS solution

Codes of Least Squares

```
function [c,r] = LeastSquaresLineFit(x,y)

% LeastSquaresLineFit    LS fit of data to  $y = c(1)*x + c(2)$ 
%
% Synopsis:    c    = LeastSquaresLineFit(x,y)
%              [c,r] = LeastSquaresLineFit(x,y)
%
% Input:    x,y = vectors of independent and dependent variables
%
% Output:    c = vector of slope, c(1), and intercept, c(2)
%            r = (optional) the sum of squared residuals
%

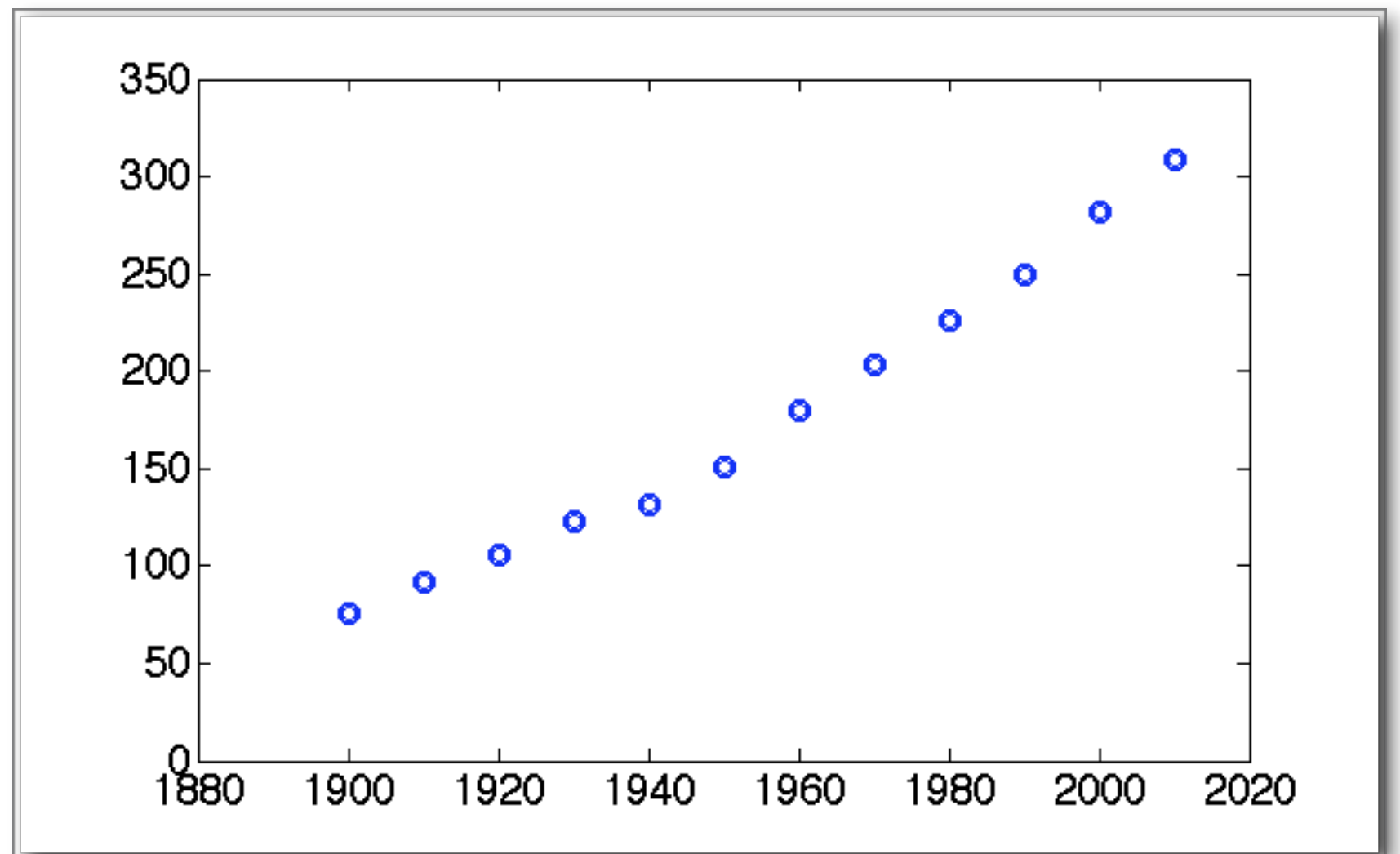
if length(y)~= length(x), error('x and y are not compatible'); end

x = x(:);  y = y(:);    % Make sure that x and y are column vectors
A = [x ones(size(x))]; % m-by-n matrix of overdetermined system
c = (A'*A)\(A'*y);      % Solve normal equations
if nargout>1
    r = norm(y - A*c)^2;% (y - A*c)'*(y - A*c)
end
```

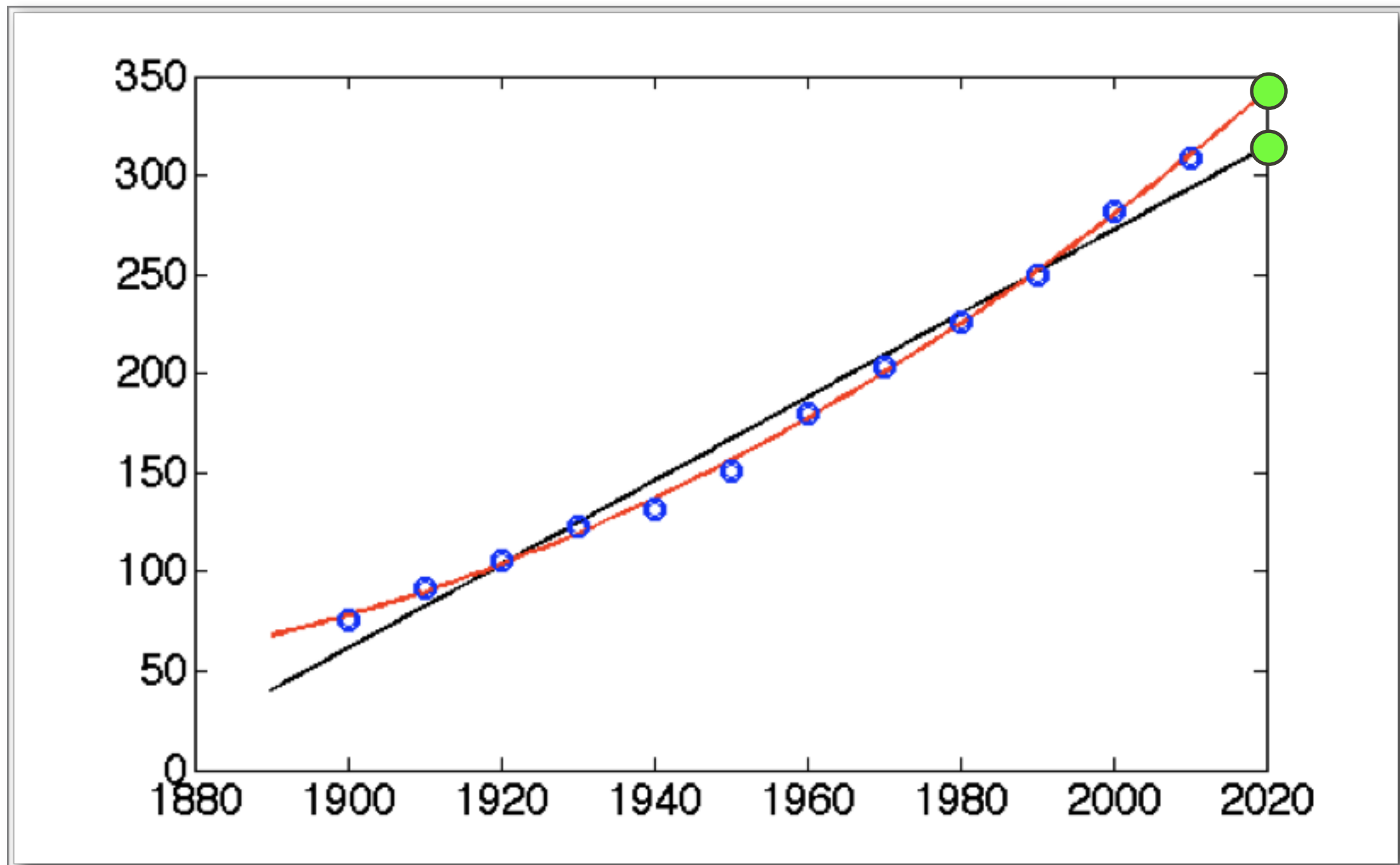
Example of Least Squares

The data are the total population (millions) of the United States for the years 1900 to 2010. Model the population growth by LS fitting and predict the population for 2020

t	y
1900	75.995
1910	91.972
1920	105.711
1930	123.203
1940	131.669
1950	150.697
1960	179.323
1970	203.212
1980	226.505
1990	249.633
2000	281.422
2010	308.748



Least Squares Line Fitting



Robust Estimation

Robust estimation

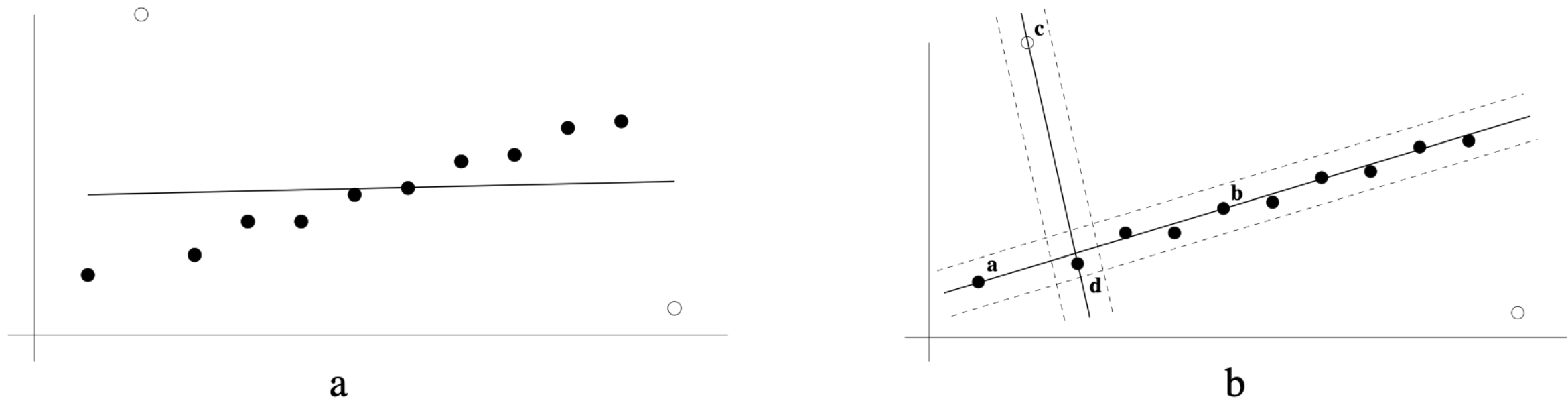


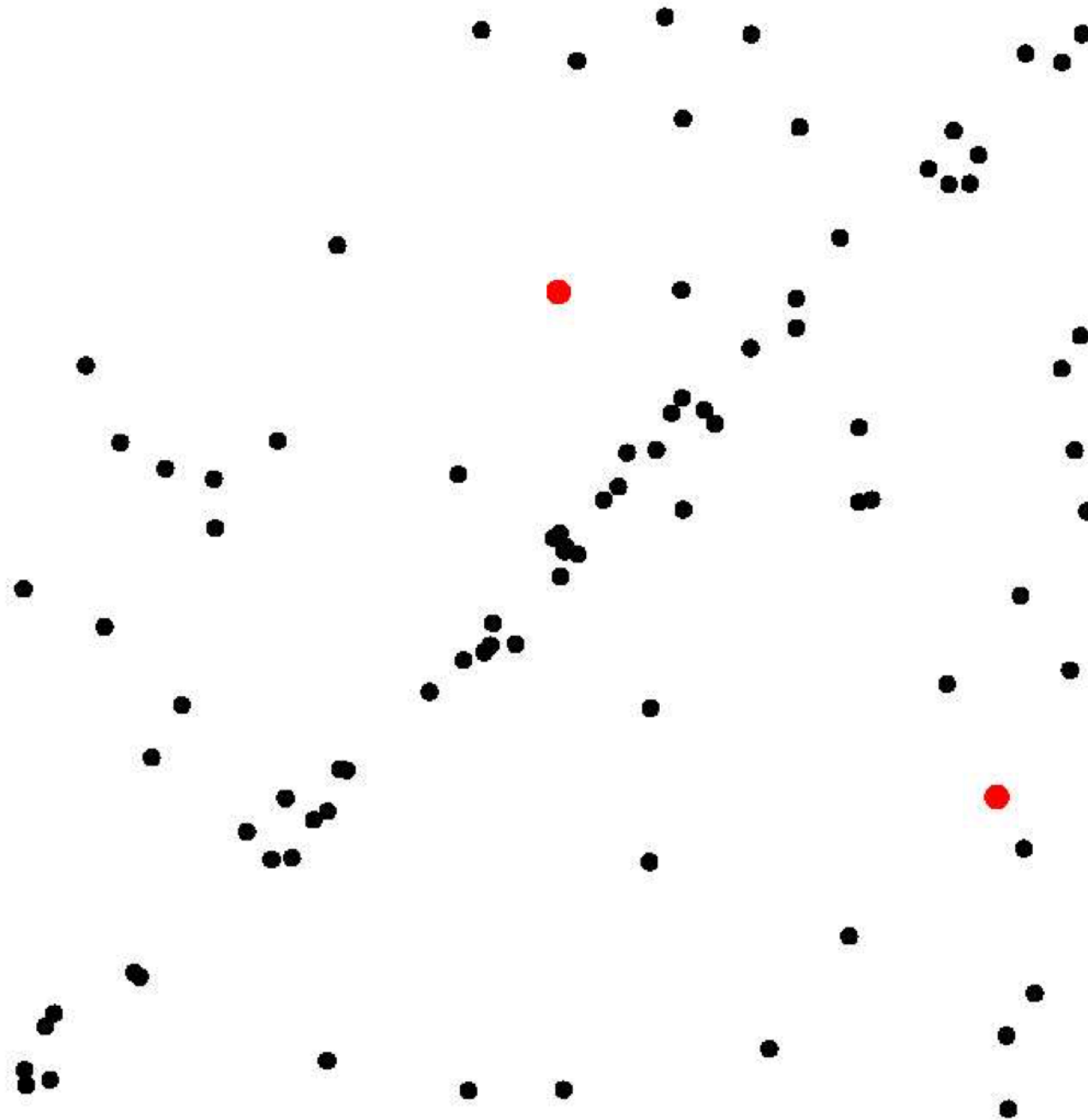
Fig. 4.7. Robust line estimation. *The solid points are inliers, the open points outliers. (a) A least-squares (orthogonal regression) fit to the point data is severely affected by the outliers. (b) In the RANSAC algorithm the support for lines through randomly selected point pairs is measured by the number of points within a threshold distance of the lines. The dotted lines indicate the threshold distance. For the lines shown the support is 10 for line $\langle \mathbf{a}, \mathbf{b} \rangle$ (where both of the points $\mathbf{a}$ and $\mathbf{b}$ are inliers); and 2 for line $\langle \mathbf{c}, \mathbf{d} \rangle$ where the point $\mathbf{c}$ is an outlier.*

RANSAC

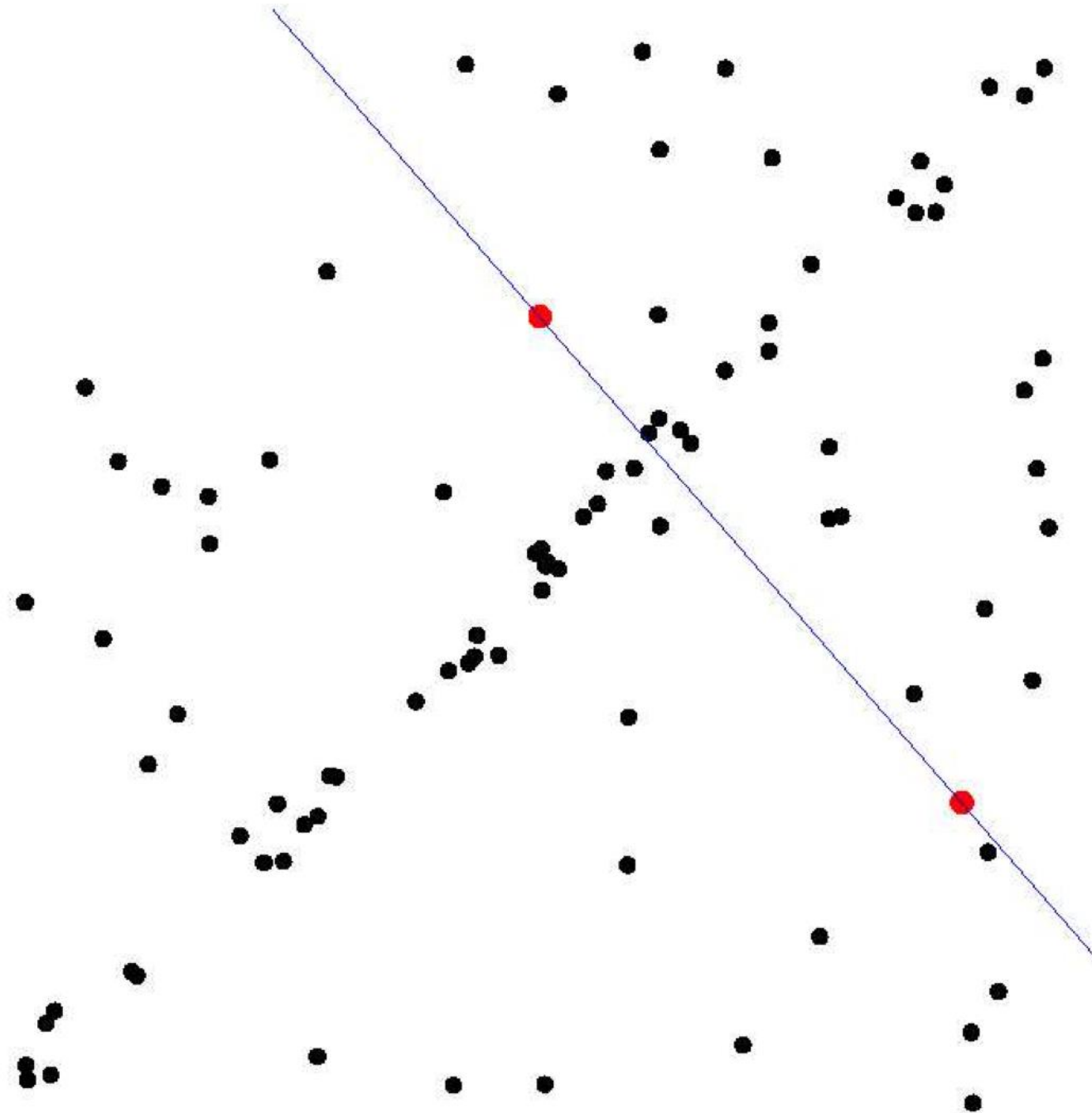


RANSAC

- **Select sample of m points at random**

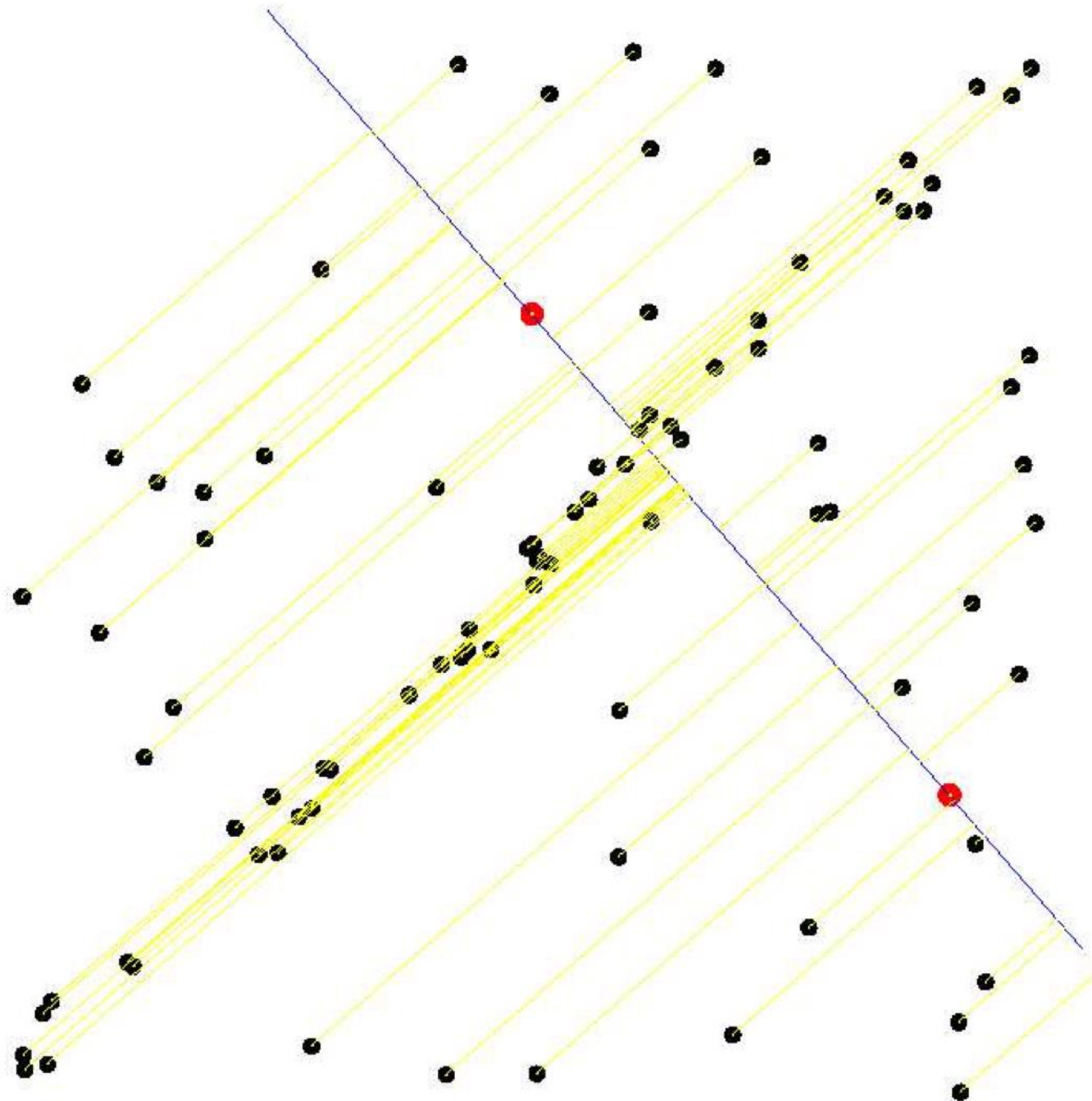


RANSAC



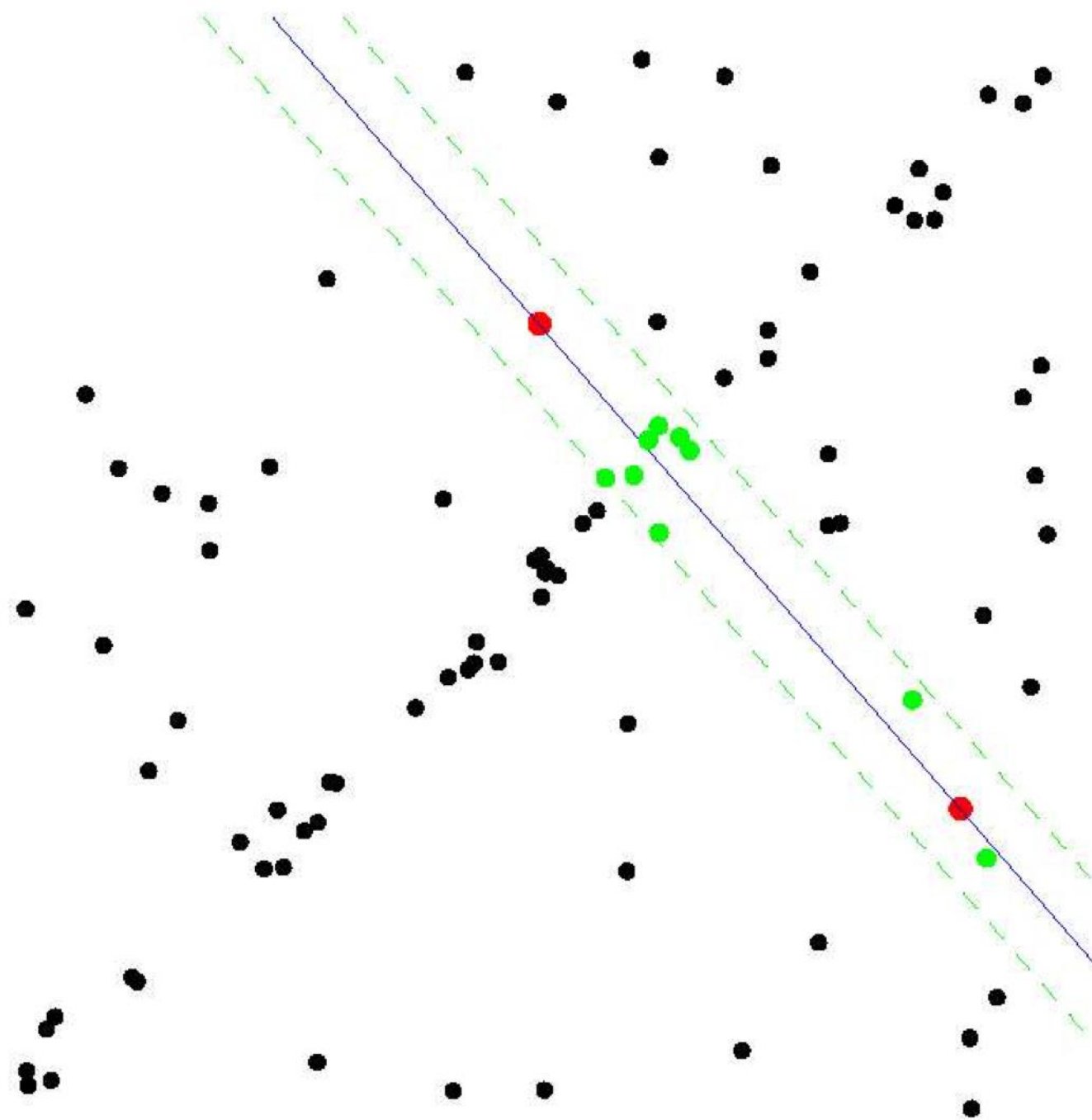
- Select sample of m points at random
- **Calculate model parameters that fit the data in the sample**

RANSAC



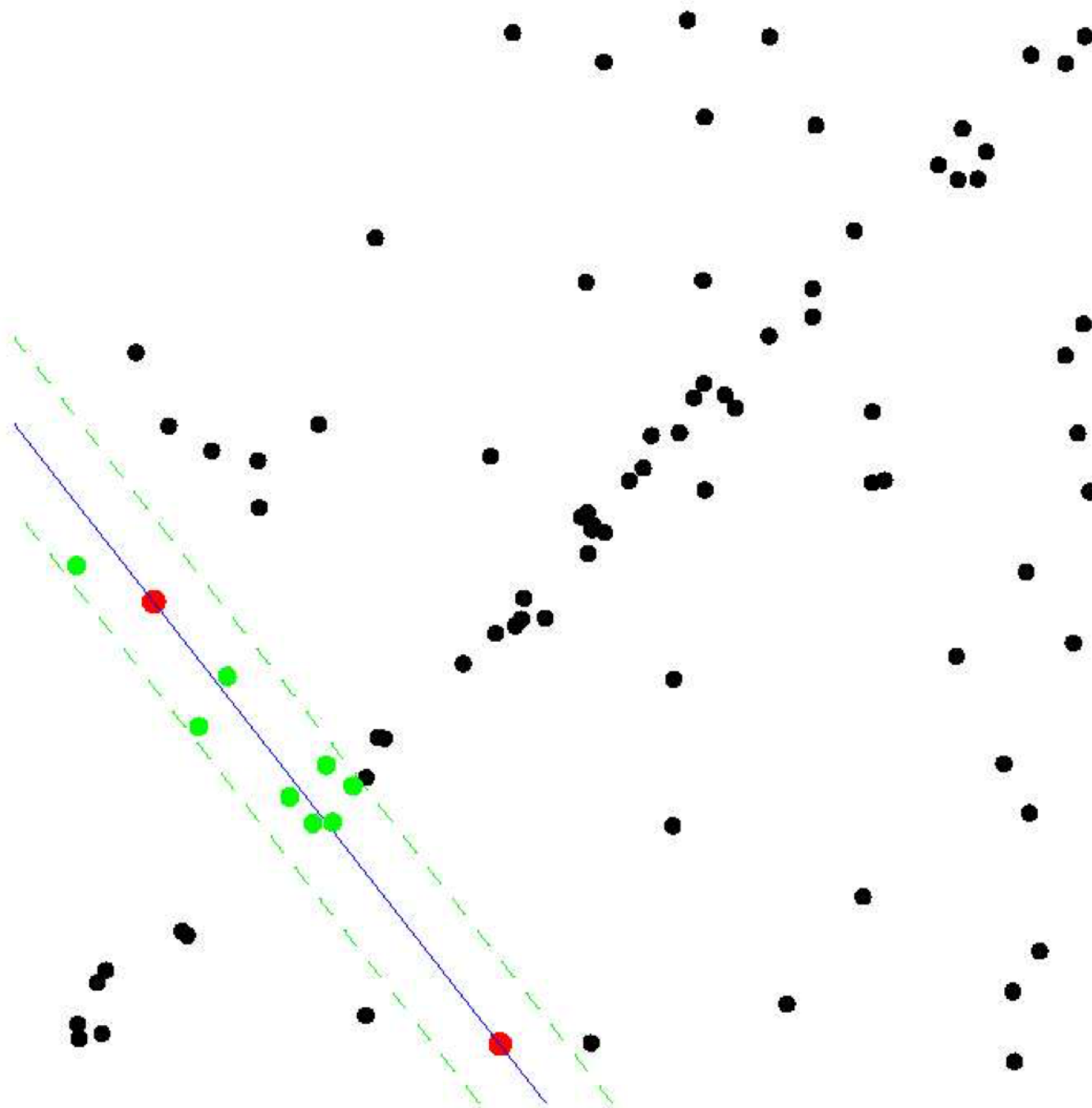
- Select sample of m points at random
- Calculate model parameters that fit the data in the sample
- **Calculate error function for each data point**

RANSAC



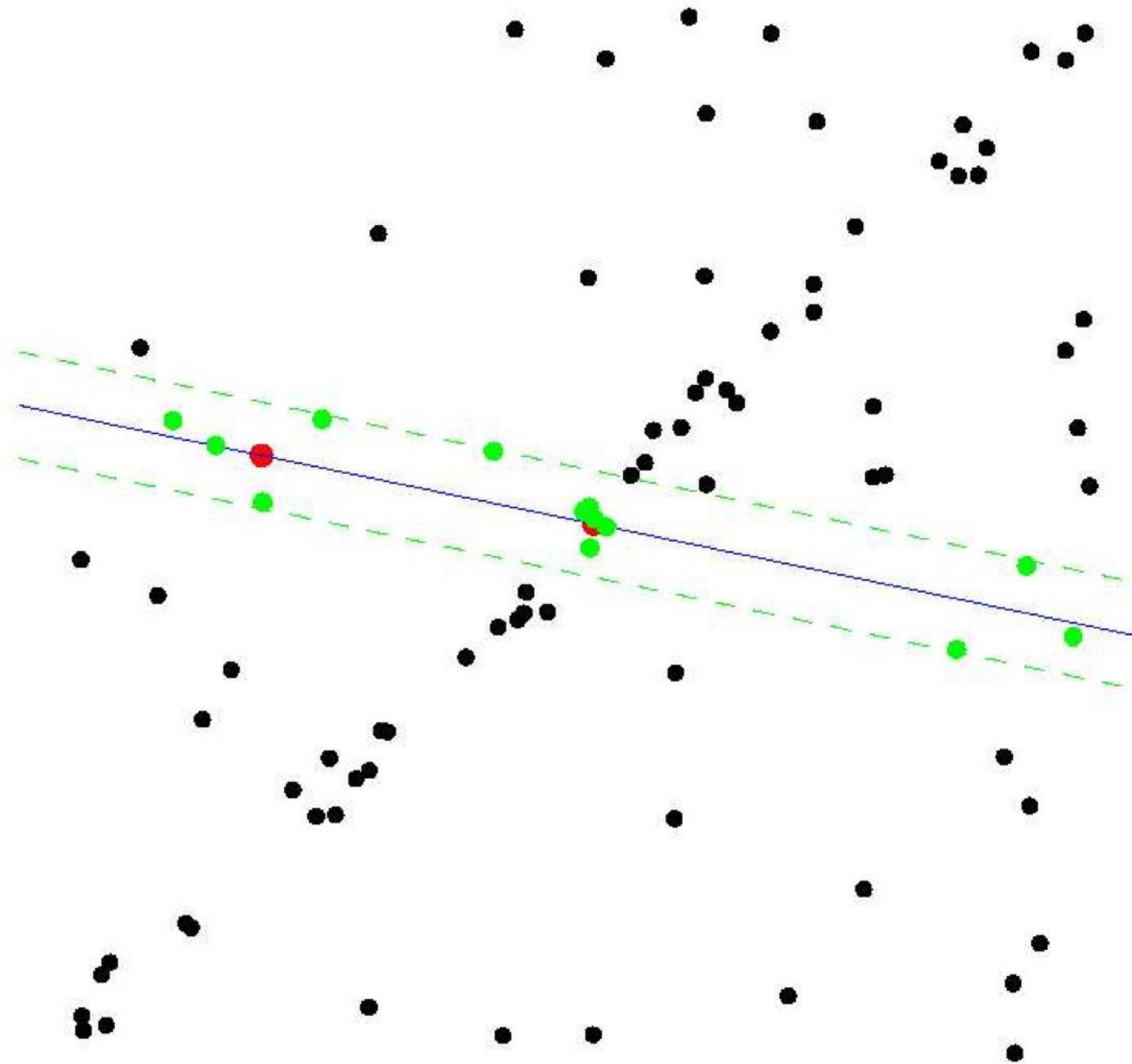
- Select sample of m points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- **Select data that support current hypothesis**

RANSAC



- Select sample of m points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- Select data that support current hypothesis
- **Repeat sampling**

RANSAC



- Select sample of m points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- Select data that support current hypothesis
- **Repeat sampling**

RANSAC

ALL-OUTLIER SAMPLE

RANSAC time complexity

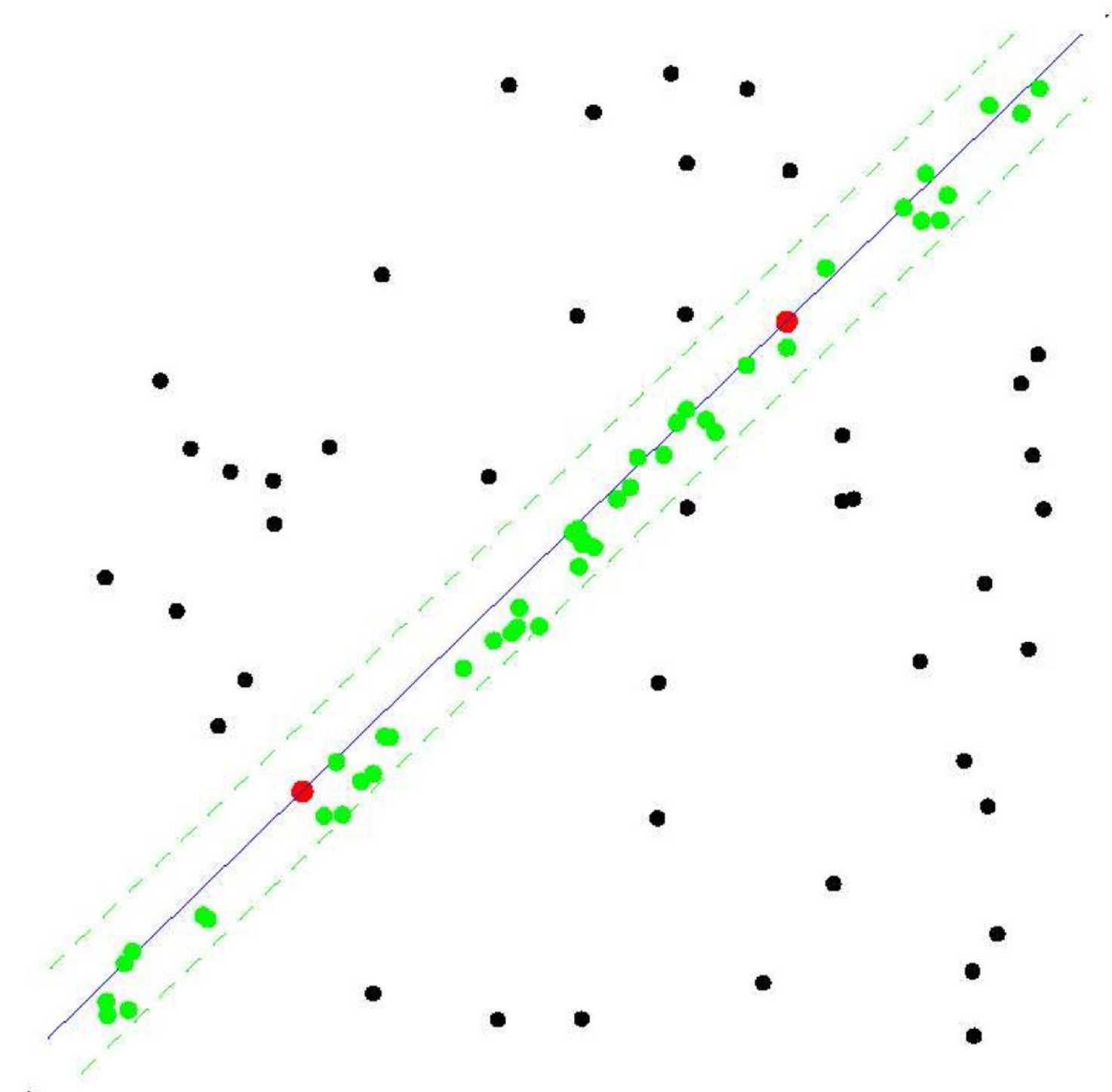
$$t = k(t_M + \bar{m}_s N)$$

k ... number of samples
drawn

N ... number of data
points

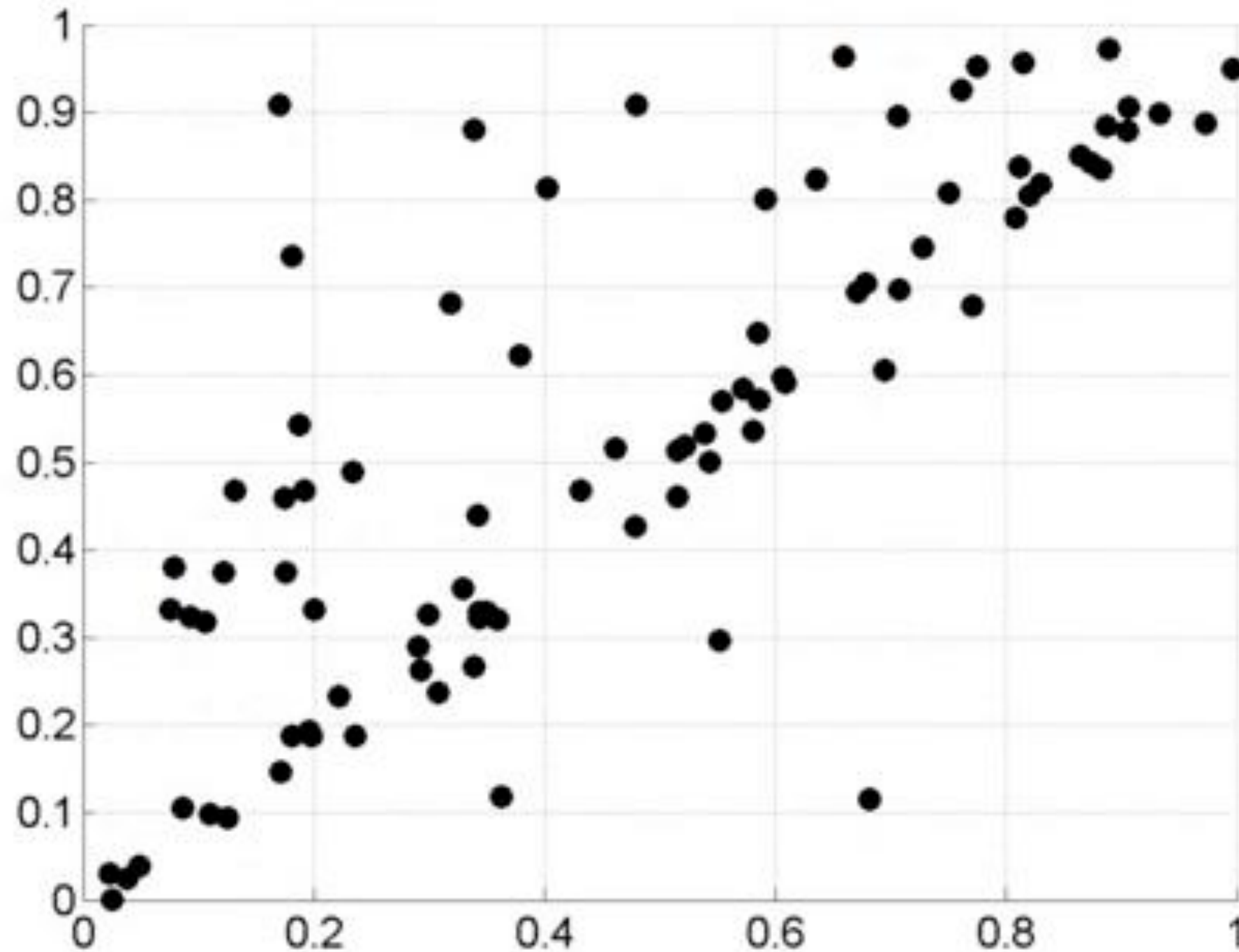
t_M ... time to compute a
single model

m_s ... average number of
models per

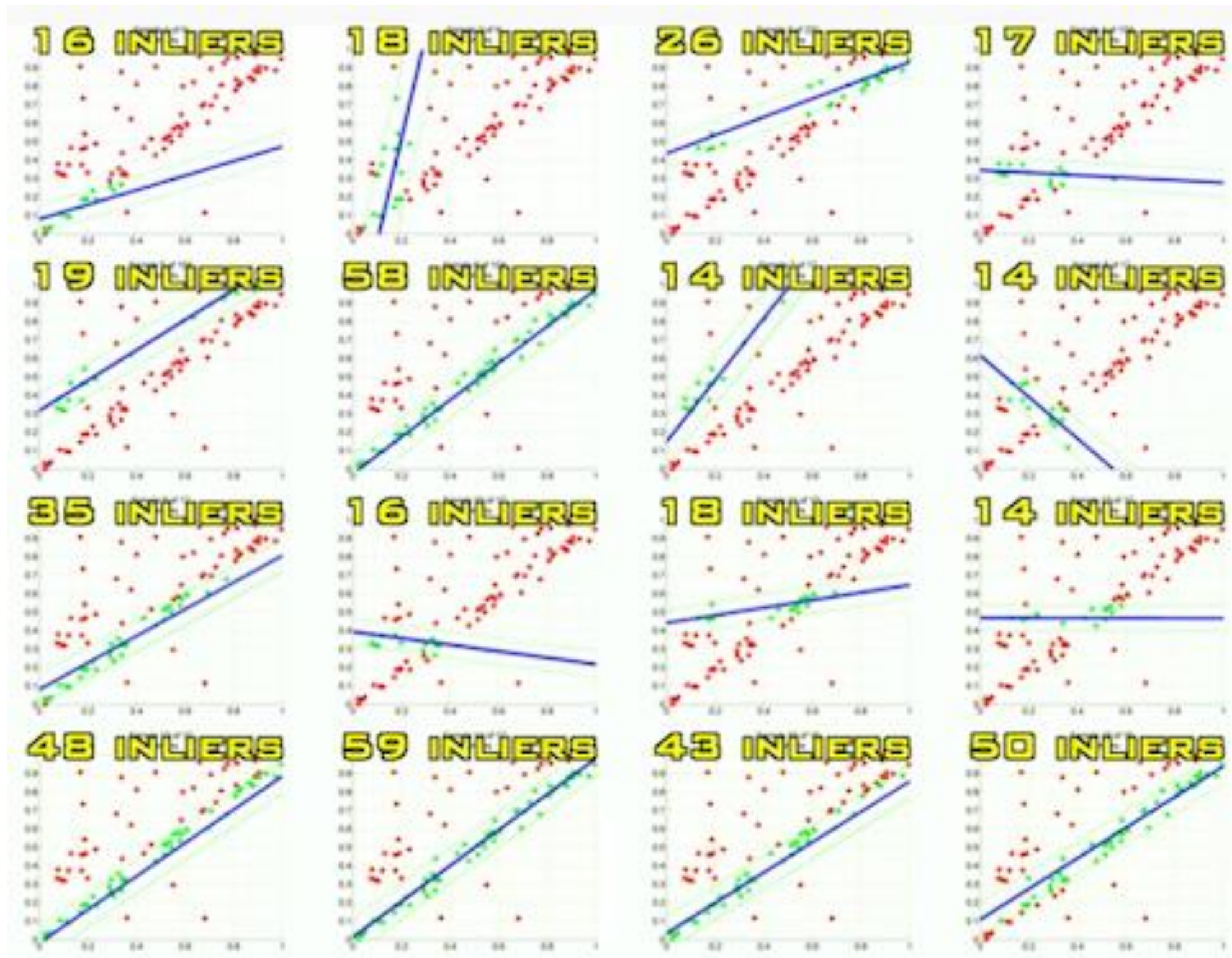


RANSAC

Sample 3 of 221



RANSAC



RANSAC

Objective

Robust fit of a model to a data set S which contains outliers.

Algorithm

- (i) Randomly select a sample of s data points from S and instantiate the model from this subset.
- (ii) Determine the set of data points S_i which are within a distance threshold t of the model. The set S_i is the consensus set of the sample and defines the inliers of S .
- (iii) If the size of S_i (the number of inliers) is greater than some threshold T , re-estimate the model using all the points in S_i and terminate.
- (iv) If the size of S_i is less than T , select a new subset and repeat the above.
- (v) After N trials the largest consensus set S_i is selected, and the model is re-estimated using all the points in the subset S_i .

RANSAC

- How to choose the distance threshold?
- How many trials?

Robust estimation

- Distance threshold

Choose a threshold t so that the probability of a point being an inlier is α (0.95)

- Choose t empirically
- Assuming zero-mean Gaussian noise with STD σ
then $d_{\perp}^2$ follows χ_m^2 distribution with m dof.

m =codimension of model

Codimension m	Model	t^2
1	line, fundamental matrix	$3.84 \sigma^2$
2	homography, camera matrix	$5.99 \sigma^2$
3	trifocal tensor	$7.81 \sigma^2$

Robust estimation

- Sample number

Choose N so that with a probability p , at least one random sample of s points is free from outliers ($p = 0.99$)

$$N = \log(1 - p) / \log(1 - (1 - \epsilon)^s)$$

Sample size	Proportion of outliers ϵ						
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177

Robust estimation

- Acceptable consensus set

Typically, terminate when the inlier ratio reaches the expected ratio of inliers

$$T = (1 - \epsilon)n$$

However, the outlier ratio is often unknown a priori.

Pick a worst case. e.g. 80% outliers yield

$$N = \log(1 - p) / \log(1 - (1 - \epsilon)^s)$$

Robust estimation

- Adaptively determining the number of samples
 - $N = \infty$, sample_count = 0.
 - While $N > \text{sample_count}$ Repeat
 - Choose a sample and count the number of inliers.
 - Set $\epsilon = 1 - (\text{number of inliers}) / (\text{total number of points})$
 - Set N from ϵ and (4.18) with $p = 0.99$.
 - Increment the sample_count by 1.
 - Terminate.

$$N = \log(1 - p) / \log(1 - (1 - \epsilon)^s)$$

Robust estimation

RANSAC

RANdom SAMple Consensus

Maximises the number of inliers according to some threshold;
sensitive to appropriate threshold selection.

MLESAC

Maximum Likelihood Estimate SAMple Consensus

Maximises the likelihood that the data was generated from the
sample-fitted model, e.g. a mixture model of inliers and outliers.

MAPSAC

Maximum *A Posteriori* SAMple Consensus

Maximises the posterior by extending MLESAC
to incorporate priors.

BOGOSAC

Bloody 'Orrible Ghastly Outrageous SAMple Confounder

Maximises randomness of results by randomly selecting the
type of model to fit to each random sample.

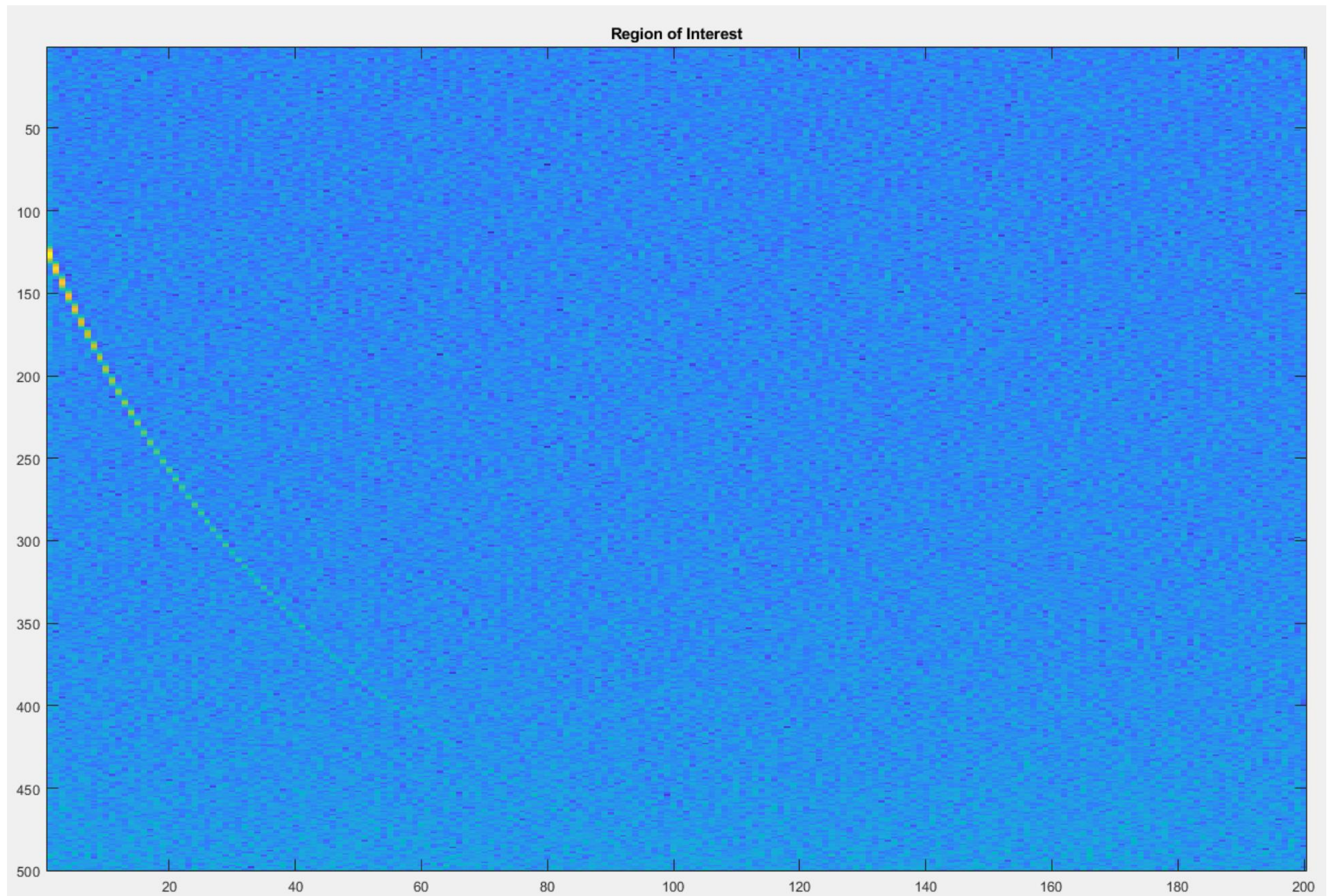
Other Robust Algorithm

- RANSAC: maximizes the number of inliers
- LMedS: minimizes the median error (no threshold, but fail with $>50\%$ outliers)
- Case deletion (small number of outliers)
- Iterative weighted LS
- Other variations

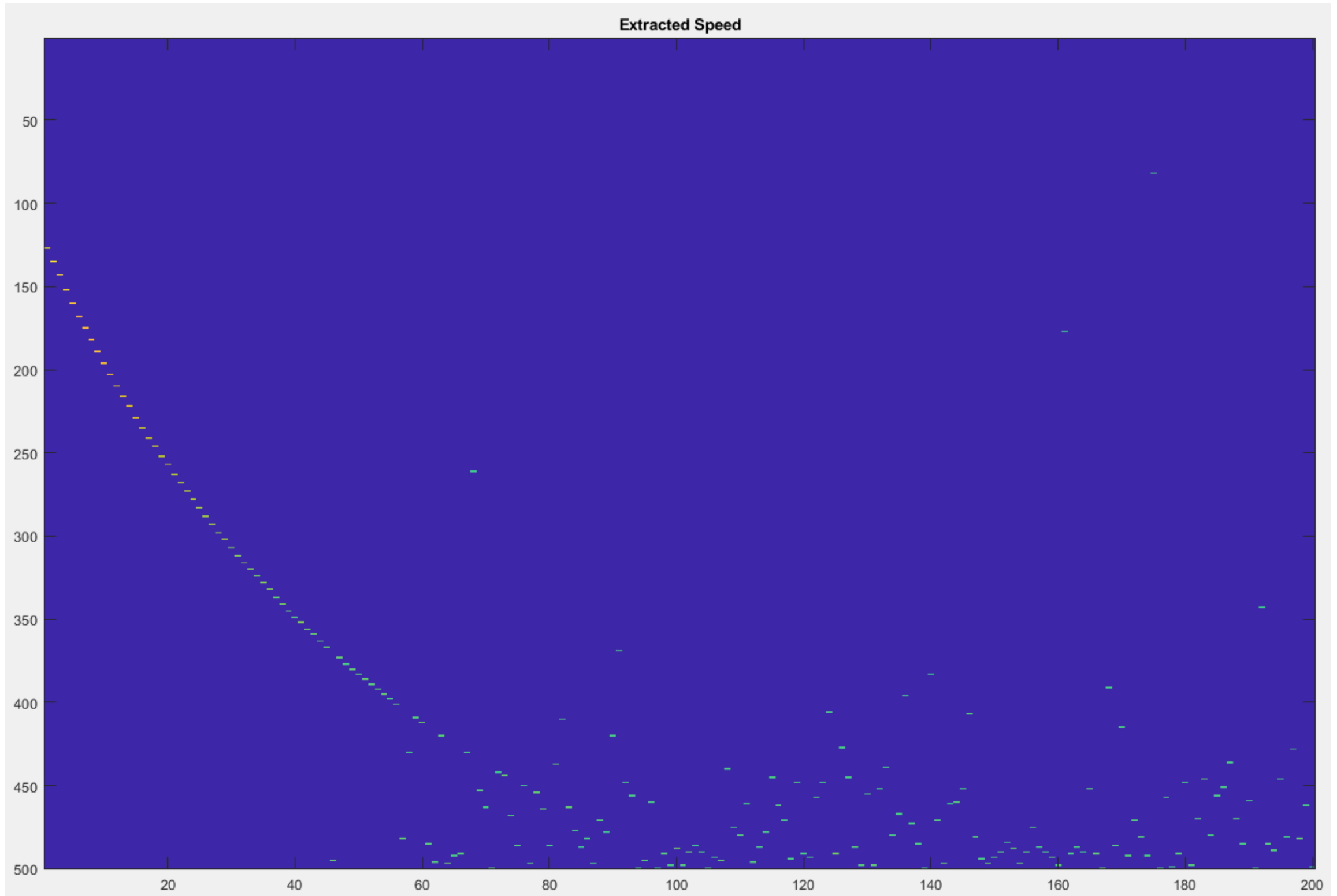
Object Detection and Recognition



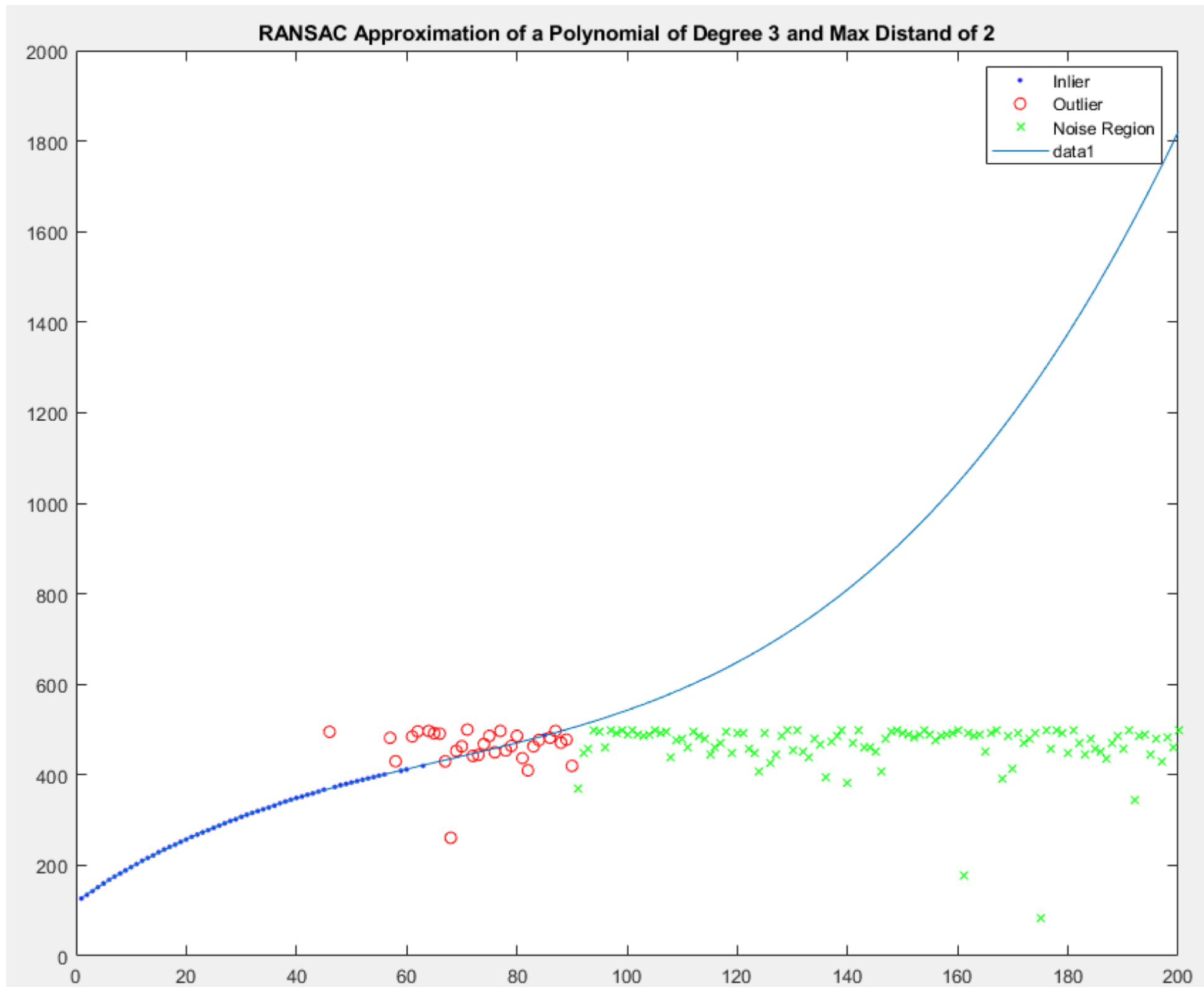
Example



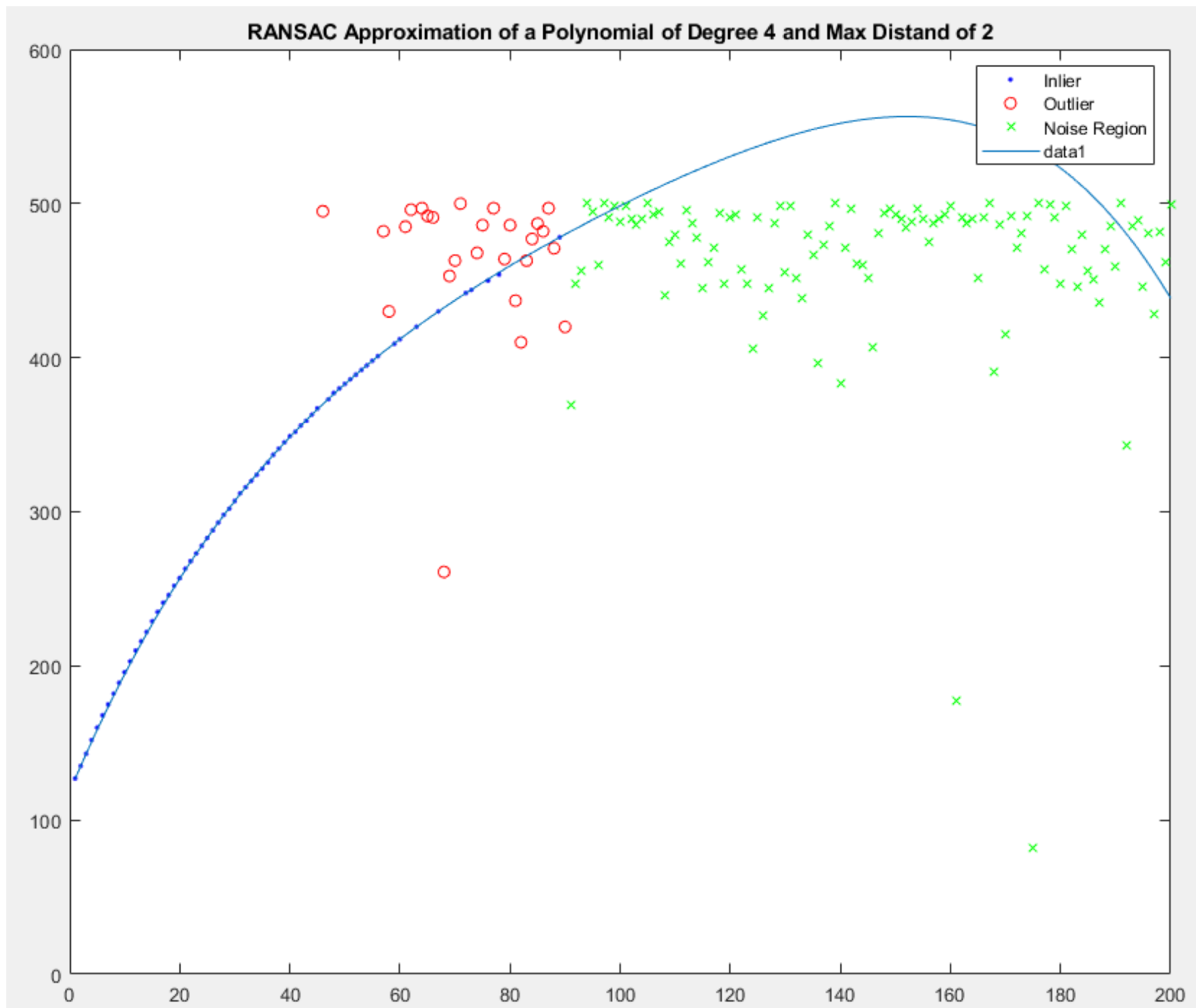
Example



Example



Example



Example

```
[P, inlierIdx] = fitPolynomialRANSAC(points(1:90, :), 4, 2);  
yRecover = polyval(P, points(:, 1));  
no_inlier = sum(inlierIdx);  
figure;  
plot(points(inlierIdx, 1), points(inlierIdx, 2), 'b.',  
      points(~inlierIdx, 1), points(~inlierIdx, 2), 'ro',  
      points(91:200, 1), points(91:200, 2), 'gx');  
legend('Inlier', 'Outlier', 'Noise Region')  
hold on;  
plot(points(:, 1), yRecover);  
title('RANSAC Approximation of a Polynomial of Degree 4  
and Max Distance of 2');
```

Application: noise removal

Read image

```
I = imread('eight.tif');  
imshow(I)
```



Application: noise removal

Add noise

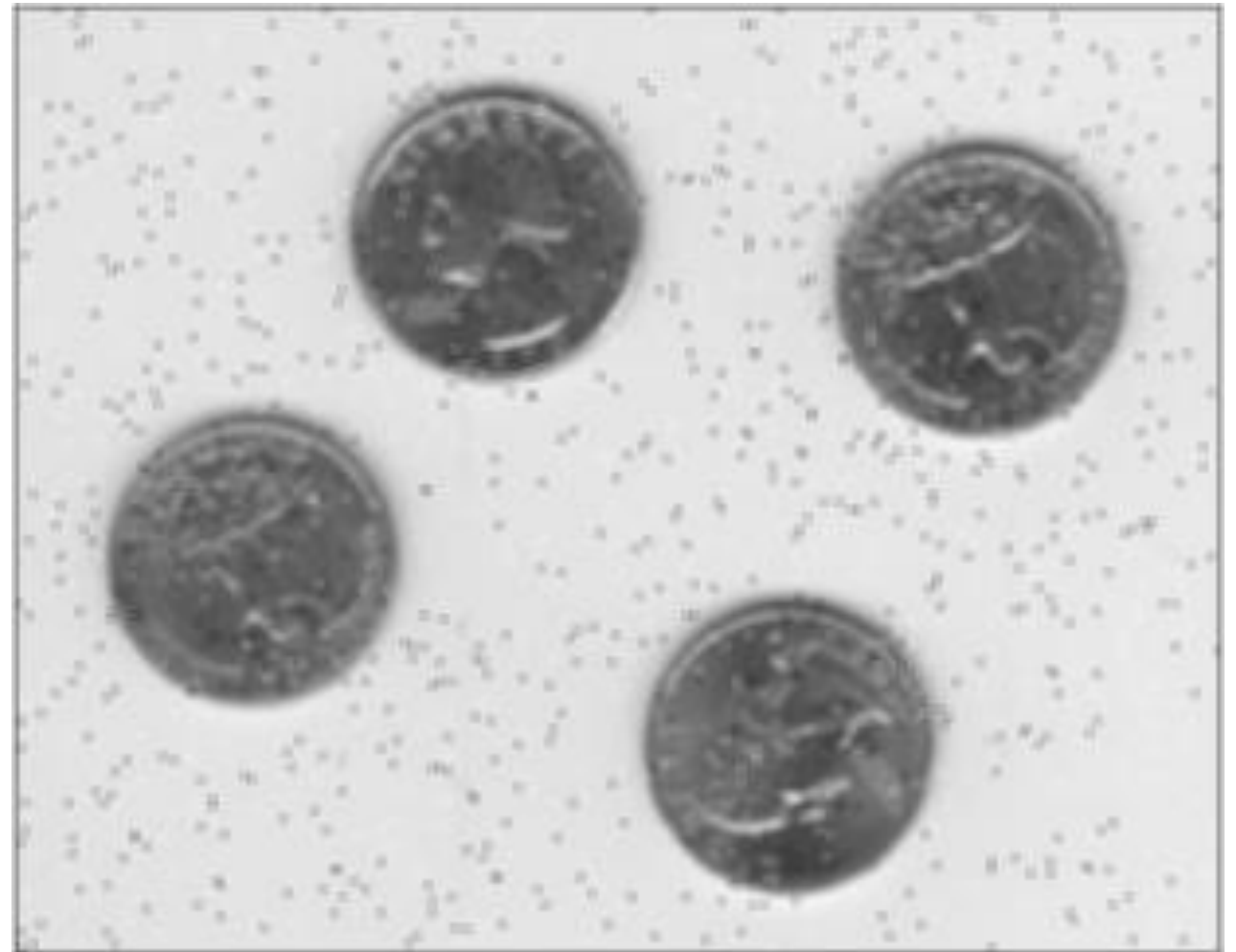
```
J = imnoise(I, 'salt & pepper', 0.02);  
figure, imshow(J)
```



Application: noise removal

Average filter

```
K = filter2(fspecial('average',3),J)/255;  
figure, imshow(K)
```



Application: noise removal

Median filter

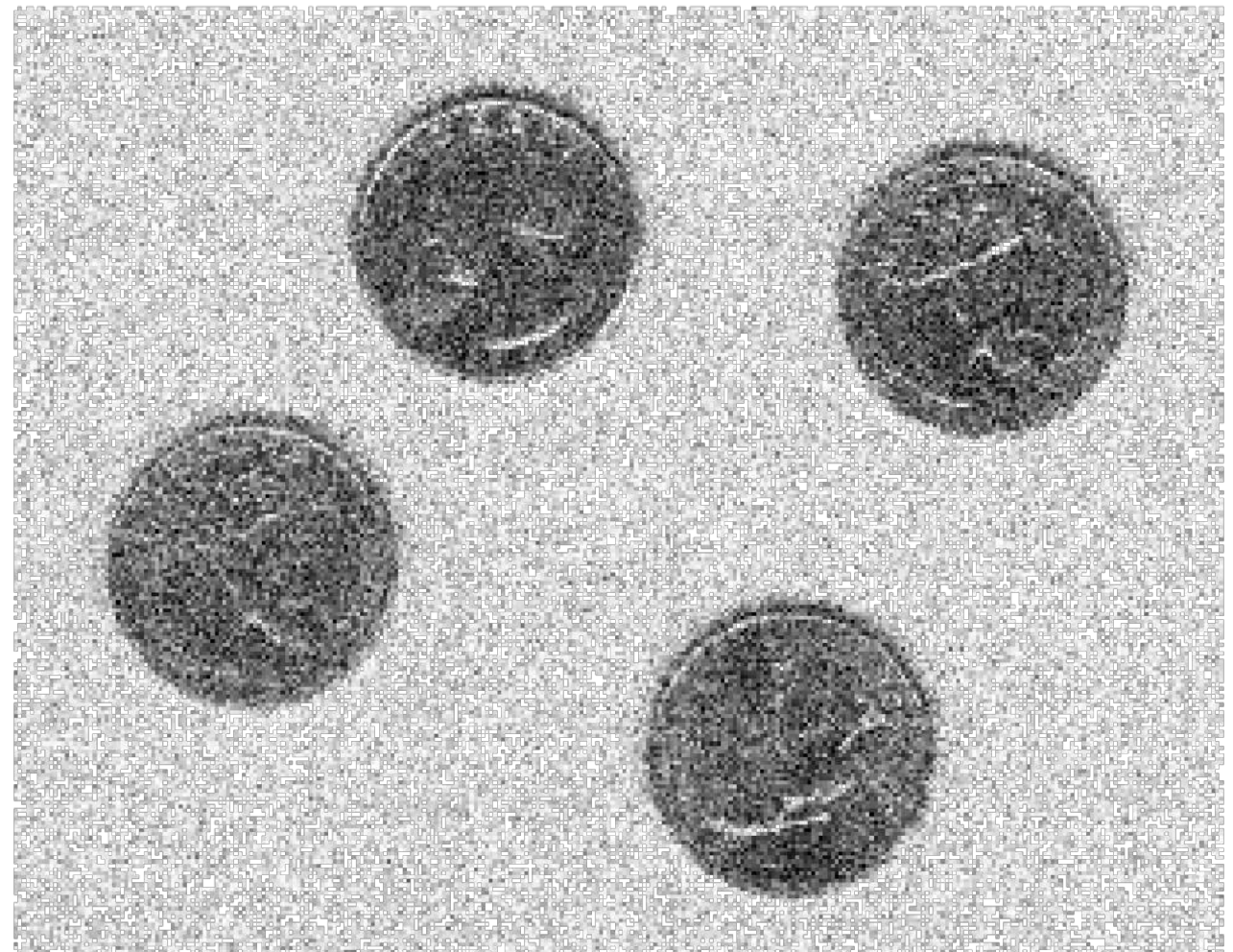
```
L = medfilt2(J,[3 3]);  
figure, imshow(L)
```



Application: noise removal

Add noise

```
J = imnoise(I, 'gaussian', 0, 0.01);  
figure, imshow(J2)
```



Application: edge detection

Read image

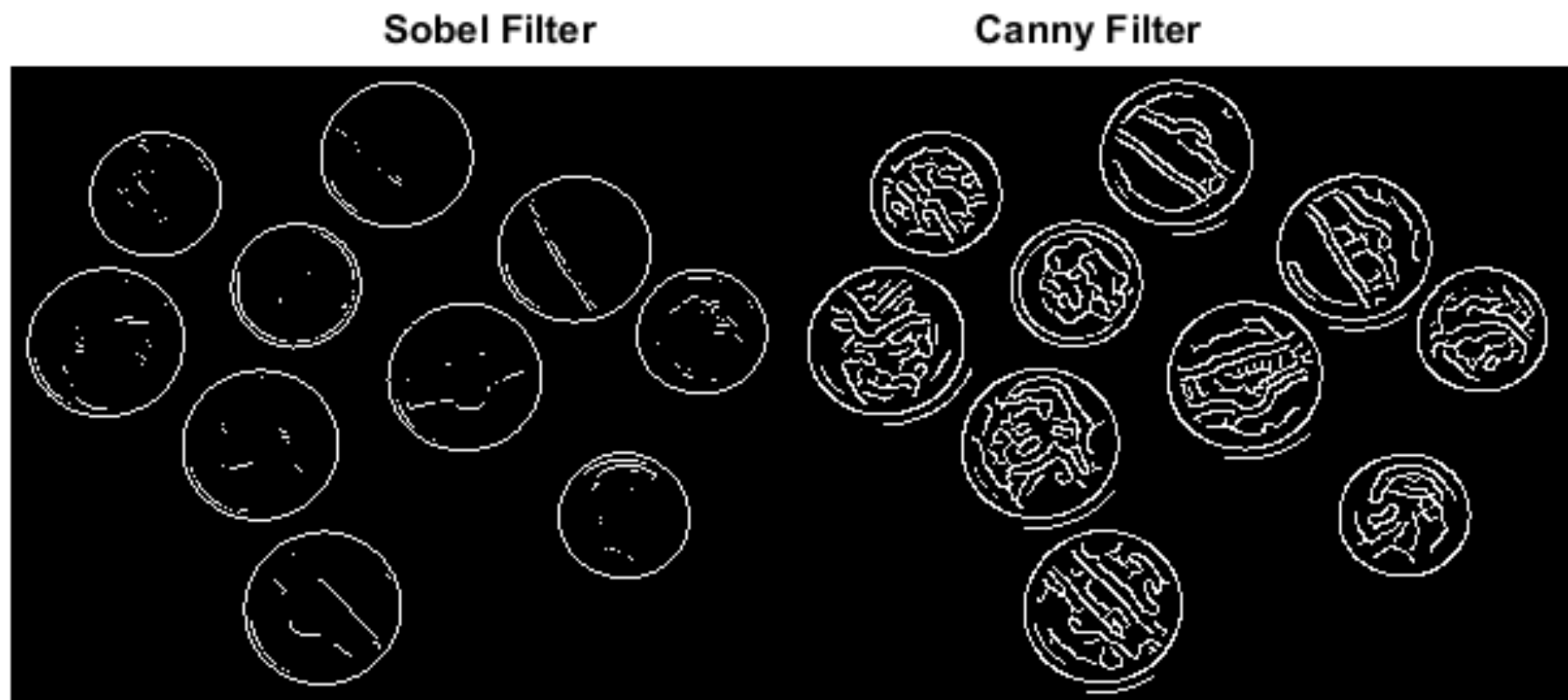
```
I = imread('coins.png');  
imshow(I)
```



Application: edge detection

Edge detection

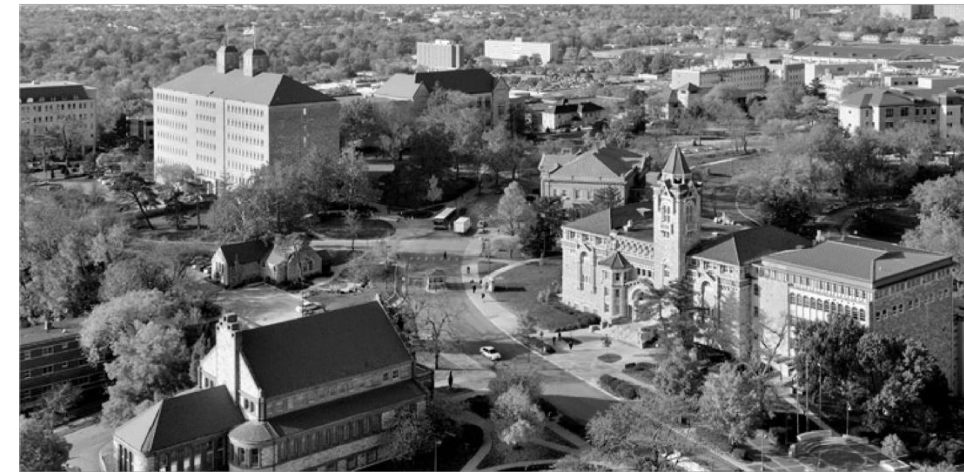
```
BW1 = edge(I, 'sobel');  
BW2 = edge(I, 'canny');  
figure;  
imshowpair(BW1, BW2, 'montage')  
title('Sobel Filter' 'Canny Filter');
```



Application: edge detection

Edge detection

```
I = rgb2gray(imread('ku.jpg'));  
imshow(I)  
BW1 = edge(I, 'sobel');  
BW2 = edge(I, 'canny');  
figure;  
imshowpair(BW1, BW2, 'montage')
```



Application: Cell Detection

Read image

```
I = imread('cell.tif');  
figure, imshow(I), title('original image');  
text(size(I,2),size(I,1)+15, ...  
      'Image courtesy of Alan Partin', ...  
      'FontSize',7,'HorizontalAlignment','right');  
text(size(I,2),size(I,1)+25, ...  
      'Johns Hopkins University', ...  
      'FontSize',7,'HorizontalAlignment','right');
```

original image



Image courtesy of Alan Partin
Johns Hopkins University

Application: Cell Detection

Detect entire cell

```
[~, threshold] = edge(I, 'sobel');  
fudgeFactor = .5;  
BWs = edge(I, 'sobel', threshold * fudgeFactor);  
figure, imshow(BWs), title('binary gradient mask');
```

binary gradient mask



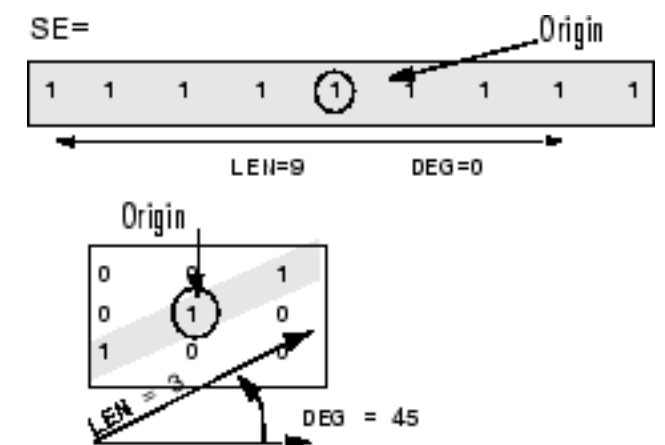
Application: Cell Detection

Detect entire cell

```
se90 = strel('line', 3, 90);      % creates a flat linear structuring element
se0 = strel('line', 3, 0);

BWsdil = imdilate(BWs, [se90 se0]);
figure, imshow(BWsdil), title('dilated gradient mask');
```

dilated gradient mask



Application: Cell Detection

Fill interior gaps

```
BWdfill = imfill(BWsdil, 'holes');  
figure, imshow(BWdfill);  
title('binary image with filled holes');
```

binary image with filled holes



Application: Cell Detection

Remove connected objects on border

```
BWnobord = imclearborder(BWdfill, 4);  
figure, imshow(BWnobord), title('cleared border image');
```

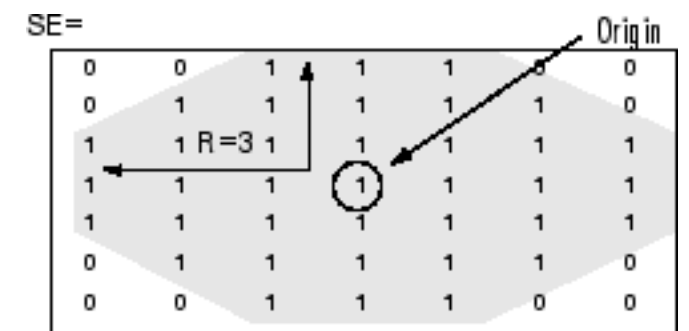
cleared border image



Application: Cell Detection

Smoothen the object

```
seD = strel('diamond',1);  
BWfinal = imerode(BWnobord,seD);  
BWfinal = imerode(BWfinal,seD);  
figure, imshow(BWfinal), title('segmented image');
```



segmented image

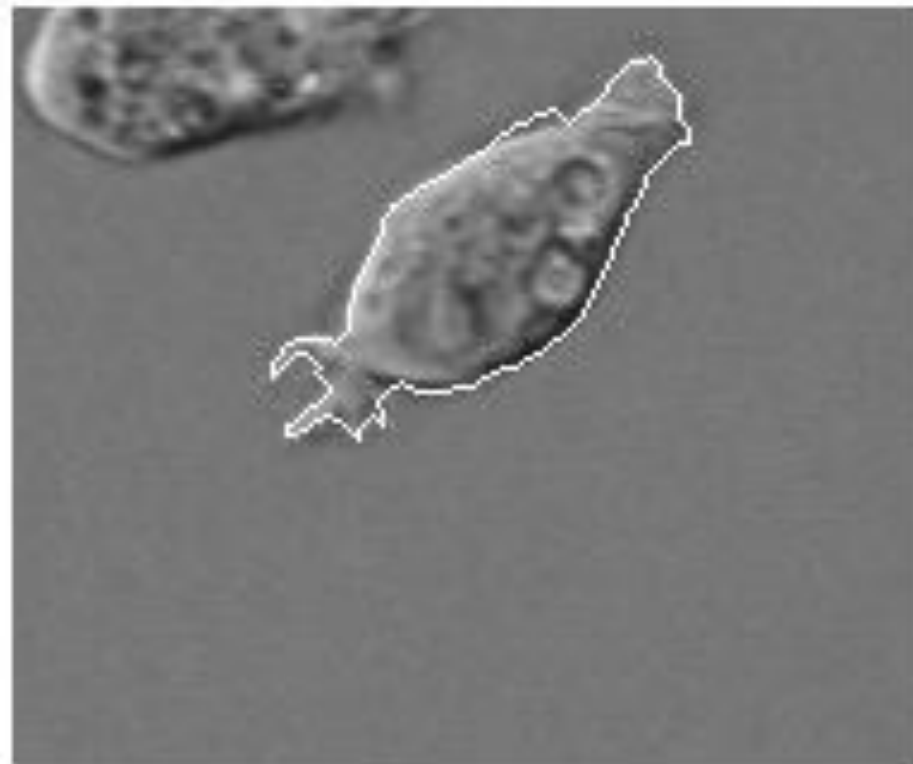


Application: Cell Detection

Smoothen the object

```
BWoutline = bwperim(BWfinal);  
Segout = I;  
Segout(BWoutline) = 255;  
figure, imshow(Segout), title('outlined original image');
```

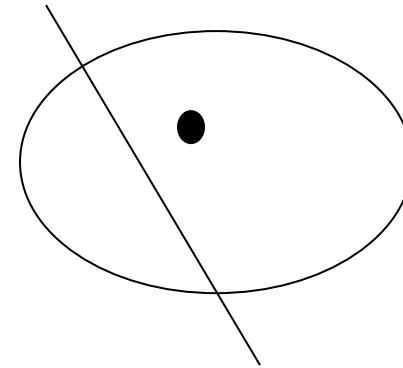
outlined original image



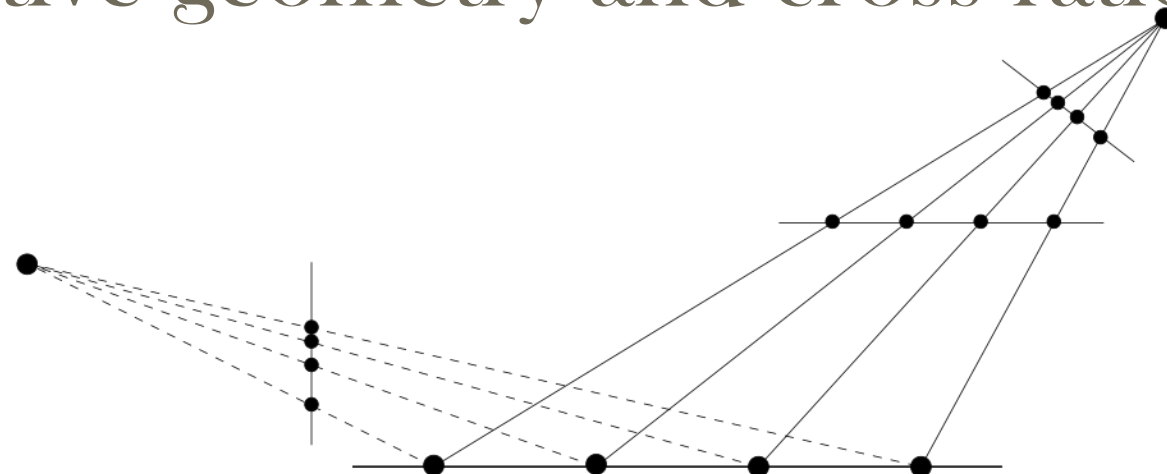
Projective Geometry

2D Projective Geometry

- Points, lines, and conics
- Transformations & invariants



- 1D projective geometry and cross-ratio



Homogeneous coordinates

- Homogeneous representation of lines

$$ax + by + c = 0 \qquad (a, b, c)^T$$

$$(ka)x + (kb)y + kc = 0, \forall k \neq 0$$

$$(a, b, c)^T \sim k(a, b, c)^T$$

- An equivalent class of vectors under this relationship is known as homogeneous vector
- The set of all equivalent classes of vectors in $\mathbb{R}^3 - (0, 0, 0)^T$ forms the 2D projective space

Homogeneous coordinates

- Homogeneous representation of points

A point $\mathbf{x} = (x, y)^T$ lies on the line $\mathbf{l} = (a, b, c)^T$

if and only if $ax + by + c = 0$.

$$(x, y, 1)(a, b, c)^T = (x, y, 1)\mathbf{l} = 0$$

The point $\mathbf{x}$ lies on the line $\mathbf{l}$ if and only if $\mathbf{x}^T\mathbf{l} = \mathbf{l}^T\mathbf{x} = 0$

homogeneous coordinates $\mathbf{x} = (x_1, x_2, x_3)^T$

inhomogeneous coordinates $(x, y)^T$. 2 d.o.f.

Homogeneous coordinates

- Intersection of lines

The intersection of two lines $\mathbf{l}$ and $\mathbf{l}'$ is the point $\mathbf{x} = \mathbf{l} \times \mathbf{l}'$.

two lines $\mathbf{l} = (a, b, c)^\top$ and $\mathbf{l}' = (a', b', c')^\top$

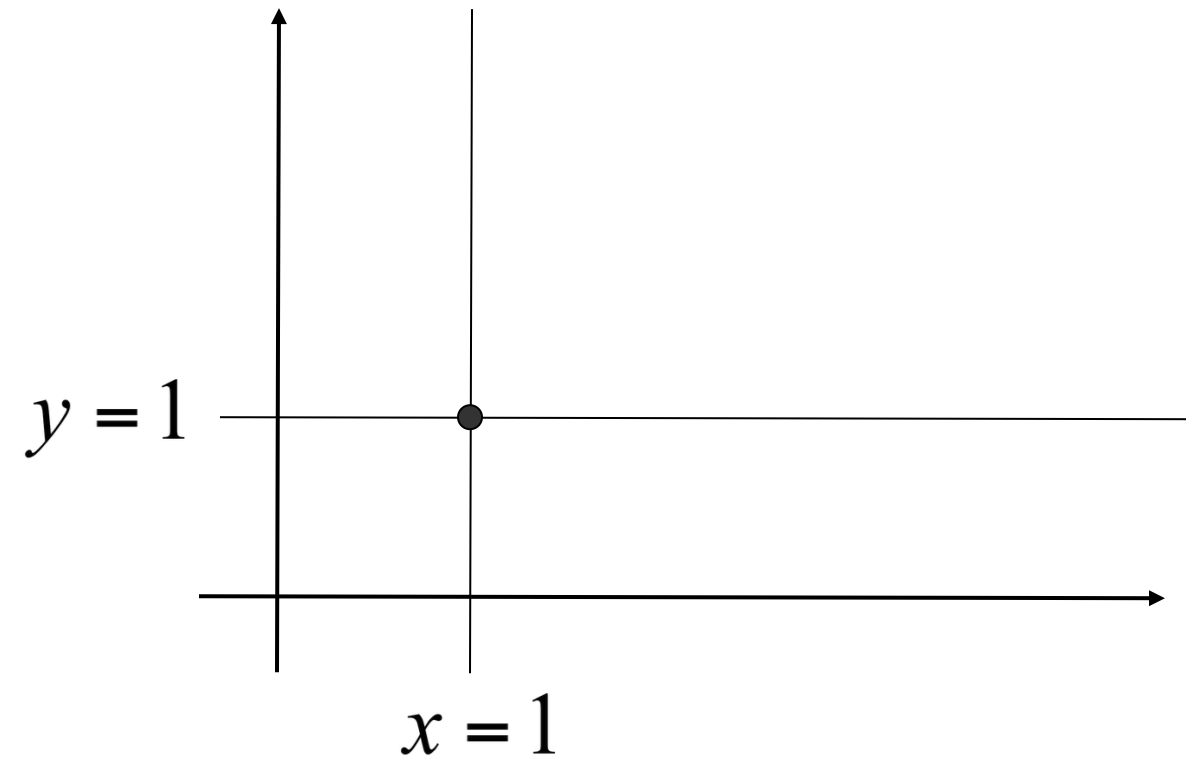
triple scalar product $\mathbf{l} \cdot (\mathbf{l} \times \mathbf{l}') = \mathbf{l}' \cdot (\mathbf{l} \times \mathbf{l}') = 0$

$$\mathbf{l}^\top \mathbf{x} = \mathbf{l}'^\top \mathbf{x} = 0$$

$\mathbf{x}$ lies on both lines.

Homogeneous coordinates

- Examples:



$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Cross product

Matrix notation [\[edit \]](#)

The cross product can also be expressed as the [formal](#) determinant:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Using [cofactor](#) expansion along the first row instead, it expands to^{[\[12\]](#)}

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= (a_2 b_3 - a_3 b_2) \mathbf{i} - (a_1 b_3 - a_3 b_1) \mathbf{j} + (a_1 b_2 - a_2 b_1) \mathbf{k}, \end{aligned}$$

which gives the components of the resulting vector directly.

Cross product

Conversion to matrix multiplication [\[edit \]](#)

The vector cross product also can be expressed as the product of a [skew-symmetric matrix](#) and a vector:

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= [\mathbf{a}]_{\times} \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \\ \mathbf{a} \times \mathbf{b} &= [\mathbf{b}]_{\times}^T \mathbf{a} = \begin{bmatrix} 0 & b_3 & -b_2 \\ -b_3 & 0 & b_1 \\ b_2 & -b_1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix},\end{aligned}$$

where superscript ^T refers to the [transpose](#) operation, and $[\mathbf{a}]_{\times}$ is defined by:

$$[\mathbf{a}]_{\times} \stackrel{\text{def}}{=} \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}.$$

Homogeneous coordinates

- Line joining points

The line through two points $\mathbf{x}$ and $\mathbf{x}'$ is $\mathbf{l} = \mathbf{x} \times \mathbf{x}'$.

Verify: the line vector is orthogonal to both points

=> the inner product of $\mathbf{l}$ and $\mathbf{x}$ ($\mathbf{x}'$) is zero

=> both points are on the line

Ideal points & line at infinity

- Intersection of parallel lines

$$l = (a, b, c)^T \text{ and } l' = (a, b, c')^T$$

$$l \times l' = (b, -a, 0)^T \quad (b/0, -a/0)^T$$

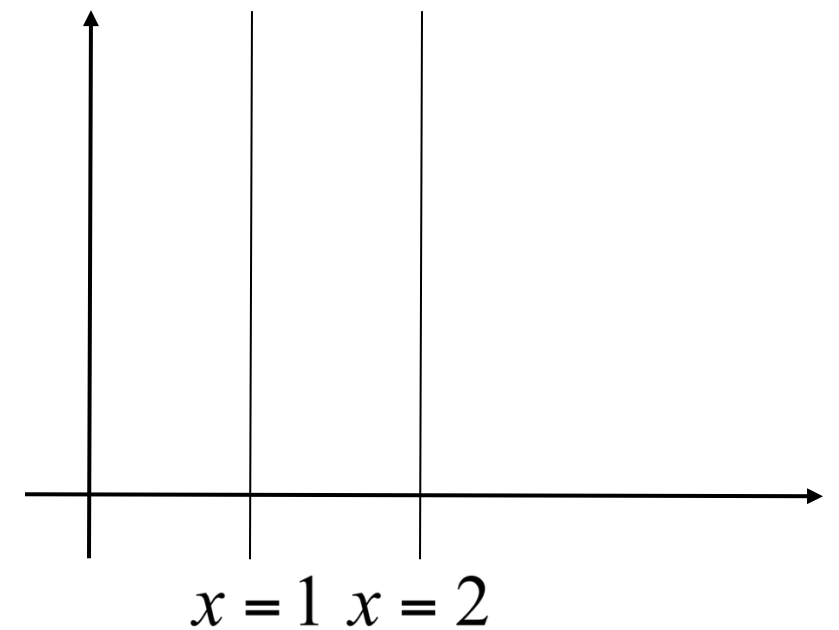
$(b, -a)$ tangent vector
 (a, b) normal direction

In general, points with homogeneous coordinates

$(x, y, 0)^T$ do not correspond to any finite point in $\mathbb{R}^2$

Homogeneous coordinates

- Examples:



$$\mathbf{x} = \mathbf{l} \times \mathbf{l}' = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 1 \\ -1 & 0 & 2 \end{vmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

the point at infinity in the direction of the y -axis.

Homogeneous coordinates

- Ideal points and line at infinity

Homogeneous vectors $\mathbf{x} = (x_1, x_2, x_3)^\top$

Ideal point, point at infinity: $(x_1, x_2, 0)^\top$

Line at infinity: $\mathbf{l}_\infty = (0, 0, 1)^\top$

Note that in $\mathbf{P}^2$ there is no distinction between ideal points and others

$$\mathbf{P}^2 = \mathbf{R}^2 \cup \mathbf{l}_\infty$$

Duality

- Duality principle:

$$\begin{array}{ccc} \mathbf{x} & \longleftrightarrow & \mathbf{l} \\ \mathbf{x}^T \mathbf{l} = 0 & \longleftrightarrow & \mathbf{l}^T \mathbf{x} = 0 \\ \mathbf{x} = \mathbf{l} \times \mathbf{l}' & \longleftrightarrow & \mathbf{l} = \mathbf{x} \times \mathbf{x}' \end{array}$$

To any theorem of 2-d projective geometry there corresponds a dual theorem, which may be derived by interchanging the role of points and lines in the original theorem

Conics and dual conics

- Conics

The equation of a conic in inhomogeneous coordinates is

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

e. a polynomial of degree 2. “Homogenizing” this by the replacements:
 $\mapsto x_1/x_3, y \mapsto x_2/x_3$ gives

$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

Matrix form:

$$\mathbf{x}^T \mathbf{C} \mathbf{x} = 0 \quad \mathbf{C} = \begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}$$

Conics and dual conics

- Five points define a conic

since if the conic passes through (x_i, y_i) then

$$ax_i^2 + bx_iy_i + cy_i^2 + dx_i + ey_i + f = 0.$$

$$\begin{pmatrix} x_i^2 & x_iy_i & y_i^2 & x_i & y_i & 1 \end{pmatrix} \mathbf{c} = 0$$

$$\mathbf{c} = (a, b, c, d, e, f)^\top$$

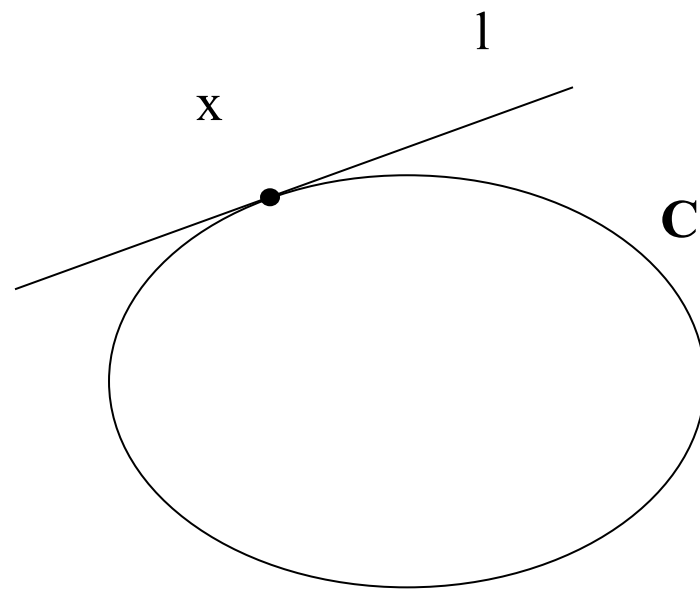
$$\begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \mathbf{c} = \mathbf{0}$$

$\mathbf{C}$ is the null vector of the
5X6 matrix

Conics and dual conics

- Tangent lines of a conic

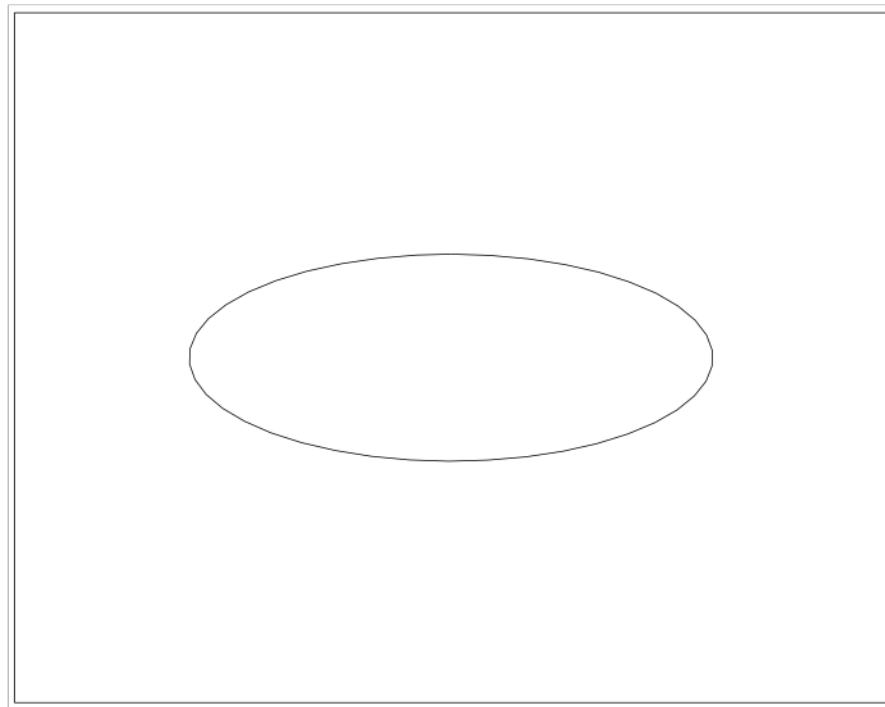
The line l tangent to C at a point x on C is given by $l = Cx$.



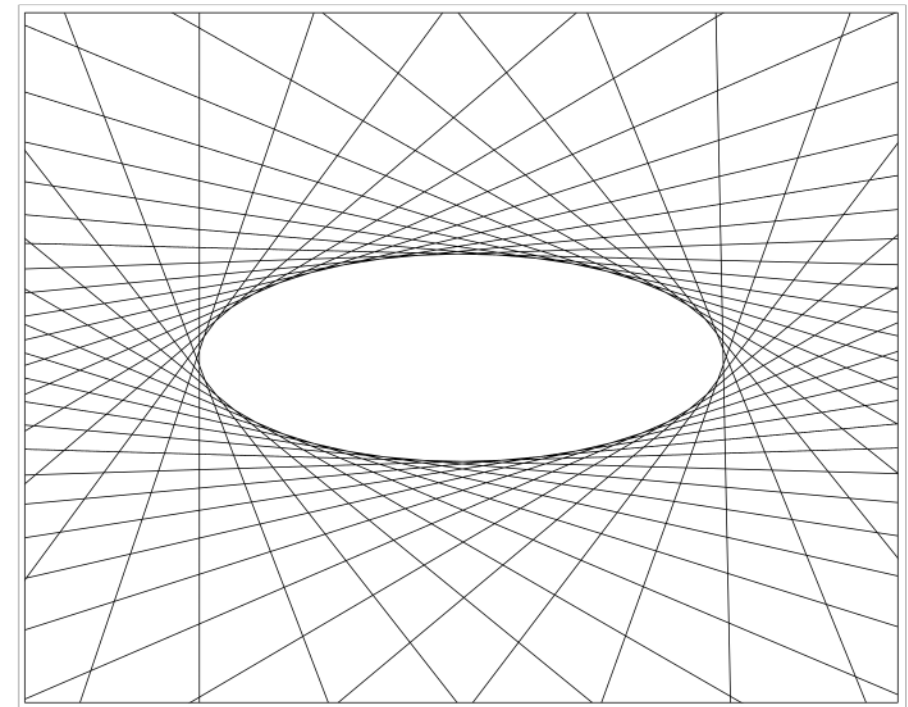
The line $l = Cx$ passes through x , since $l^T x = x^T Cx = 0$.

Conics and dual conics

- Dual conics Dual conics = line conics = conic envelopes



a



b

Fig. 2.2. (a) Points $\mathbf{x}$ satisfying $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$ lie on a point conic. (b) Lines $\mathbf{l}$ satisfying $\mathbf{l}^T \mathbf{C}^* \mathbf{l} = 0$ are tangent to the point conic $\mathbf{C}$. The conic $\mathbf{C}$ is the envelope of the lines $\mathbf{l}$.

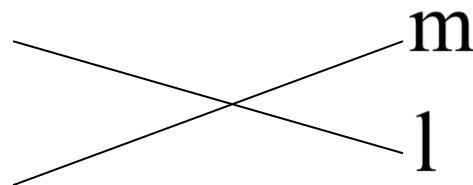
In general ($\mathbf{C}$ full rank): $\mathbf{C}^* = \mathbf{C}^{-1}$

Conics and dual conics

- Degenerate conics: A conic is degenerate if matrix $\mathbf{C}$ is not of full rank

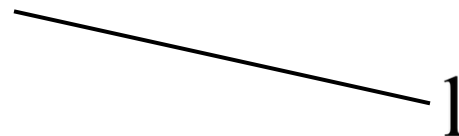
e.g. two lines (rank 2)

$$\mathbf{C} = \mathbf{l}\mathbf{m}^T + \mathbf{m}\mathbf{l}^T$$



e.g. repeated line (rank 1)

$$\mathbf{C} = \mathbf{l}\mathbf{l}^T$$



Degenerate line conics: 2 points (rank 2), double point (rank 1)

$$\mathbf{C}^* = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$$

Projective transformation

- Definition: Projective transformation

Definition 2.11. Projective transformation. A planar projective transformation is a linear transformation on homogeneous 3-vectors represented by a non-singular 3×3 matrix:

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad (2.5)$$

or more briefly, $\mathbf{x}' = \mathbf{H}\mathbf{x}$.

8 d.o.f.

Projective transformation

- Definition:

A *projectivity* is an invertible mapping h from P^2 to itself such that three points x_1, x_2, x_3 lie on the same line if and only if $h(x_1), h(x_2), h(x_3)$ do.

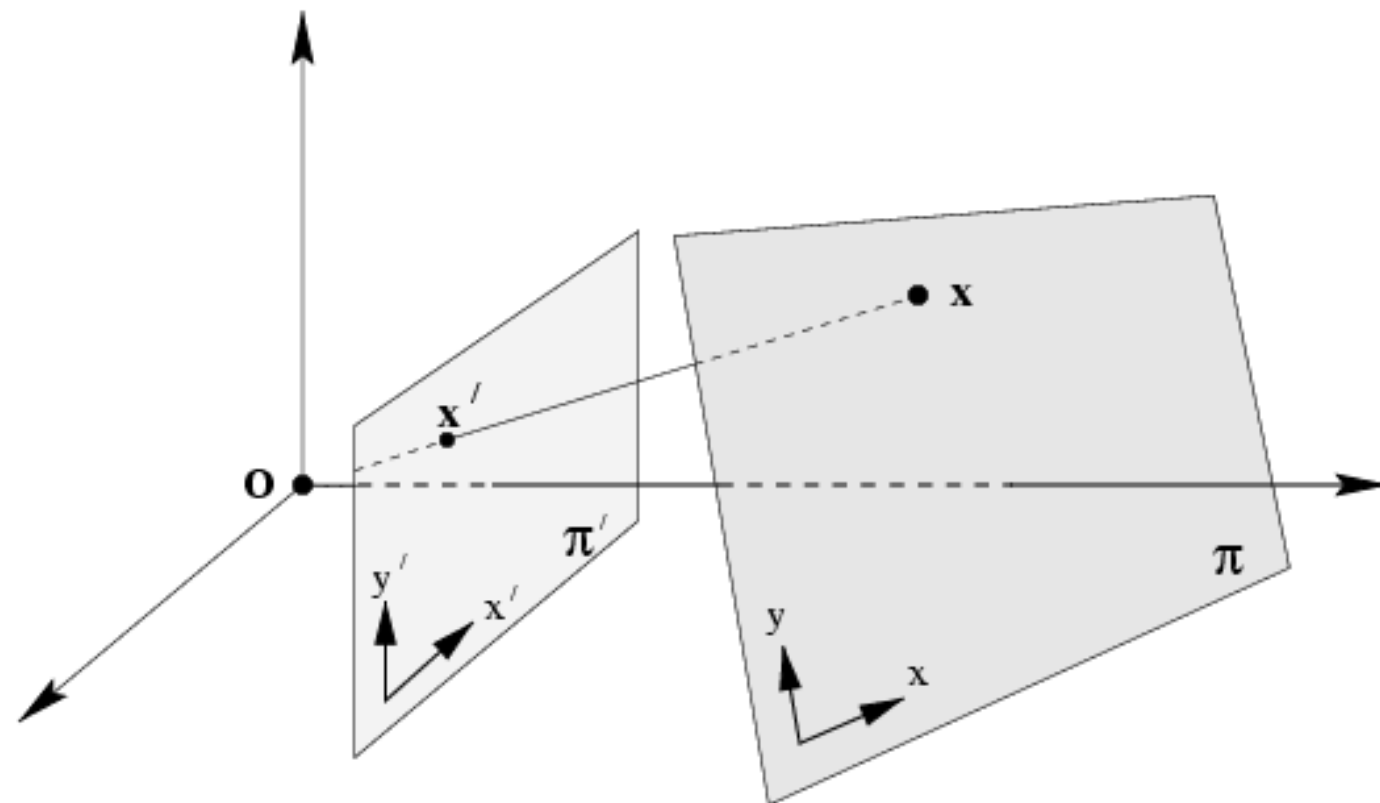
projectivity = collineation = projective transformation = homography

- Theorem:

A mapping $h:P^2 \rightarrow P^2$ is a projectivity if and only if there exist a non-singular 3×3 matrix $\mathbf{H}$ such that for any point in P^2 represented by a vector $\mathbf{x}$ it is true that $h(\mathbf{x}) = \mathbf{H}\mathbf{x}$

Projective transformation

- Mapping between planes:



central projection may be expressed by $x' = Hx$

Projective transformation

- Removing projective distortion:



select four points in a plane with know coordinates

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{x'_2}{x'_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

Projective transformation

- Removing projective distortion:

$$x'(h_{31}x + h_{32}y + h_{33}) = h_{11}x + h_{12}y + h_{13}$$

$$y'(h_{31}x + h_{32}y + h_{33}) = h_{21}x + h_{22}y + h_{23}$$

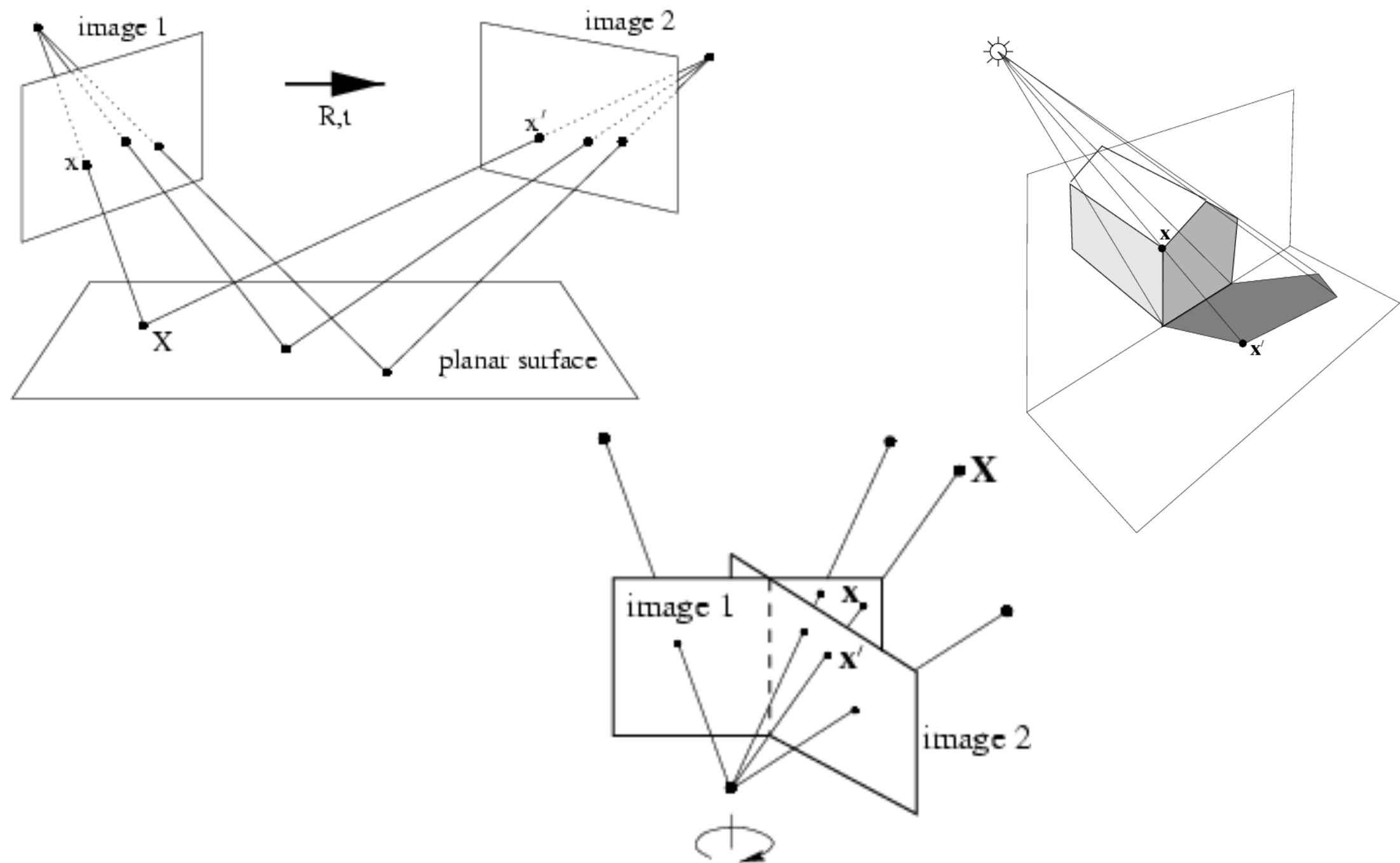
(linear in h_{ij})

(2 constraints/point, 8 dof $\Rightarrow$ 4 points needed)

Remark: no calibration at all,
better ways to compute (come later)

Projective transformation

- More examples:



transformation of lines

- Transformation of lines:

Under point transformation $\mathbf{x}'_i = \mathbf{H}\mathbf{x}_i$

A line is transformed to $\mathbf{l}' = \mathbf{H}^{-\top}\mathbf{l}$

$$\mathbf{l}'^{\top}\mathbf{x}'_i = \mathbf{l}^{\top}\mathbf{H}^{-1}\mathbf{H}\mathbf{x}_i = 0$$

$$\mathbf{l}'^{\top} = \mathbf{l}^{\top}\mathbf{H}^{-1}$$

transformation of Conics

- Transformation of conics:

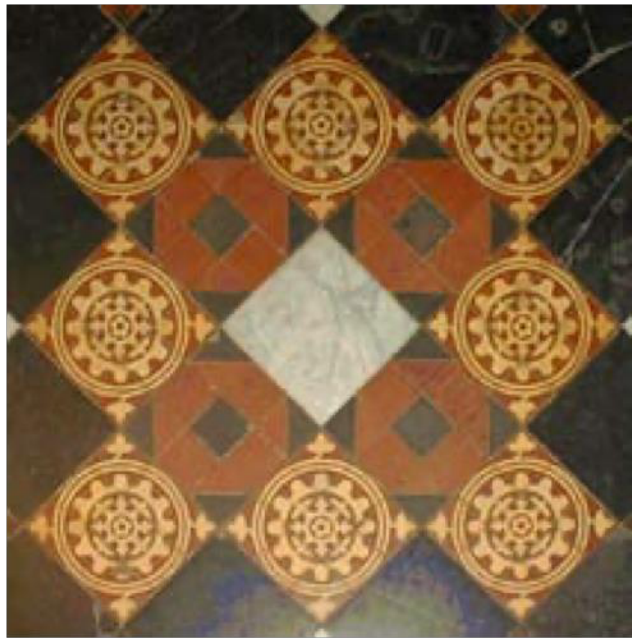
Under point transformation $\mathbf{x}'_i = \mathbf{H}\mathbf{x}_i$

A conic is transformed to $\mathbf{C}' = \mathbf{H}^{-\top}\mathbf{C}\mathbf{H}^{-1}$

$$\begin{aligned}\mathbf{x}^{\top}\mathbf{C}\mathbf{x} &= \mathbf{x}'^{\top}[\mathbf{H}^{-1}]^{\top}\mathbf{C}\mathbf{H}^{-1}\mathbf{x}' \\ &= \mathbf{x}'^{\top}\mathbf{H}^{-\top}\mathbf{C}\mathbf{H}^{-1}\mathbf{x}'\end{aligned}$$

A dual conic is transformed to $\mathbf{C}^{*'} = \mathbf{H}\mathbf{C}^{*}\mathbf{H}^{\top}$

A Hierarchy of transformations



a



b



c

Fig. 2.6. **Distortions arising under central projection.** *Images of a tiled floor. (a) **Similarity**: the circular pattern is imaged as a circle. A square tile is imaged as a square. Lines which are parallel or perpendicular have the same relative orientation in the image. (b) **Affine**: The circle is imaged as an ellipse. Orthogonal world lines are not imaged as orthogonal lines. However, the sides of the square tiles, which are parallel in the world are parallel in the image. (c) **Projective**: Parallel world lines are imaged as converging lines. Tiles closer to the camera have a larger image than those further away.*

Class I: Isometries

- *iso*=same, *metric*=measure

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \varepsilon \cos \theta & -\sin \theta & t_x \\ \varepsilon \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \quad \varepsilon = \pm 1$$

orientation preserving: $\varepsilon = 1$

orientation reversing: $\varepsilon = -1$

Class I: Isometries

- Block form:
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \varepsilon \cos \theta & -\sin \theta & t_x \\ \varepsilon \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$\mathbf{x}' = \mathbf{H}_E \mathbf{x} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x} \quad \mathbf{R}^\top \mathbf{R} = \mathbf{R} \mathbf{R}^\top = \mathbf{I}$$

3 d.o.f. (1 rotation, 2 translations)

special cases: pure rotation, pure translation

Invariants: lengths, angles, areas

Class II: Similarity

- *isometry + scale*

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} s \cos \theta & -s \sin \theta & t_x \\ s \sin \theta & s \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$\mathbf{x}' = \mathbf{H}_s \mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x} \quad s - \text{isotropic scaling}$$

4 d.o.f. (1 scale, 1 rotation, 2 translations)

also known as *equi-form* (shape preserving)
metric structure = structure up to similarity

Invariants: ratios of lengths, angles, ratios of areas, parallel lines

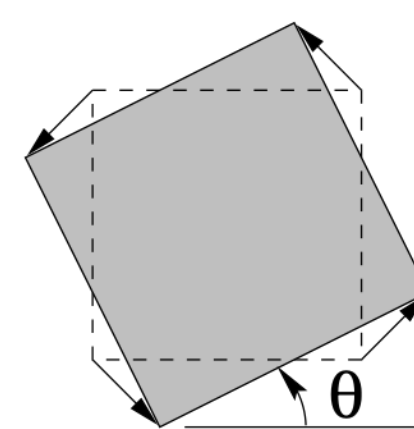
Class III: Affine transformations

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

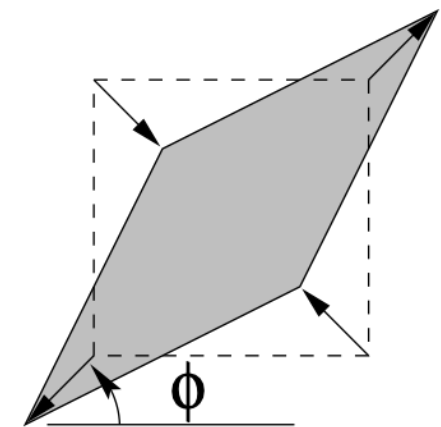
$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \mathbf{x}$$

$$\mathbf{A} = \mathbf{R}(\theta) \mathbf{R}(-\phi) \mathbf{D} \mathbf{R}(\phi)$$

$$\mathbf{U} \mathbf{D} \mathbf{V}^T = (\mathbf{U} \mathbf{V}^T) (\mathbf{V} \mathbf{D} \mathbf{V}^T) = \mathbf{R}(\theta) (\mathbf{R}(-\phi) \mathbf{D} \mathbf{R}(\phi)) \quad \mathbf{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$



rotation



deformation

6 d.o.f. (2 scales, 2 rotations, 2 translations)

non-isotropic scaling (2 extra dof: scale ratio and orientation)

Invariants: parallel lines, ratios of areas, ratio of lengths on parallel lines

Class IV: Projective transformation

- General linear transformation

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\mathbf{x}' = \mathbf{H}_P \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix} \mathbf{x} \quad \mathbf{v} = (v_1, v_2)^\top$$

8 d.o.f.

Computed from 4 point correspondences with no 3 collinear on either plane

Invariants: cross ratio of 4 points on a line (ratio of ratios)

Action on line at infinity

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ 0 \end{pmatrix} \quad \mathbf{l}' = \mathbf{H}^{-\top} \mathbf{l}$$

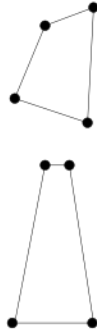
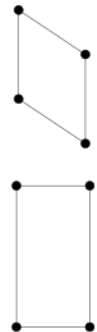
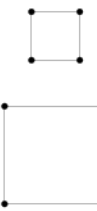
Ideal point maps to ideal point, line at infinity stays at infinity, but points move along line

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ v_1 x_1 + v_2 x_2 \end{pmatrix}$$

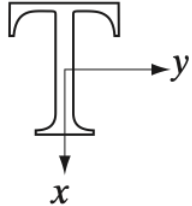
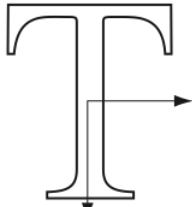
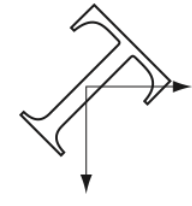
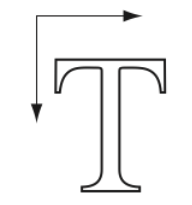
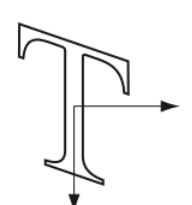
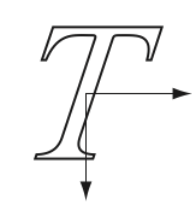
Ideal point becomes finite point,
line at infinity becomes finite line,
allows to observe vanishing points, horizon,



Overview of transformations

Group	Matrix	Distortion	Invariant properties
Projective 8 dof	$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$		Concurrency, collinearity, order of contact : intersection (1 pt contact); tangency (2 pt contact); inflections (3 pt contact with line); tangent discontinuities and cusps. cross ratio (ratio of ratio of lengths).
Affine 6 dof	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Parallelism, ratio of areas, ratio of lengths on collinear or parallel lines (e.g. midpoints), linear combinations of vectors (e.g. centroids). The line at infinity, l_∞ .
Similarity 4 dof	$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Ratio of lengths, angle. The circular points, I, J (see section 2.7.3).
Euclidean 3 dof	$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$		Length, area

Example Affine transformations

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= w \end{aligned}$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= c_x v \\ y &= c_y w \end{aligned}$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \cos \theta - w \sin \theta \\ y &= v \sin \theta + w \cos \theta \end{aligned}$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$\begin{aligned} x &= v + t_x \\ y &= w + t_y \end{aligned}$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v + s_v w \\ y &= w \end{aligned}$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= s_h v + w \end{aligned}$	

Projective geometry of 1D

$(x_1, x_2)^\top$ — 1D point of a line

$x_2 = 0$ — ideal point of a line

$\bar{\mathbf{X}}' = \mathbf{H}_{2 \times 2} \bar{\mathbf{X}}$ 1D transformation — 3 dof

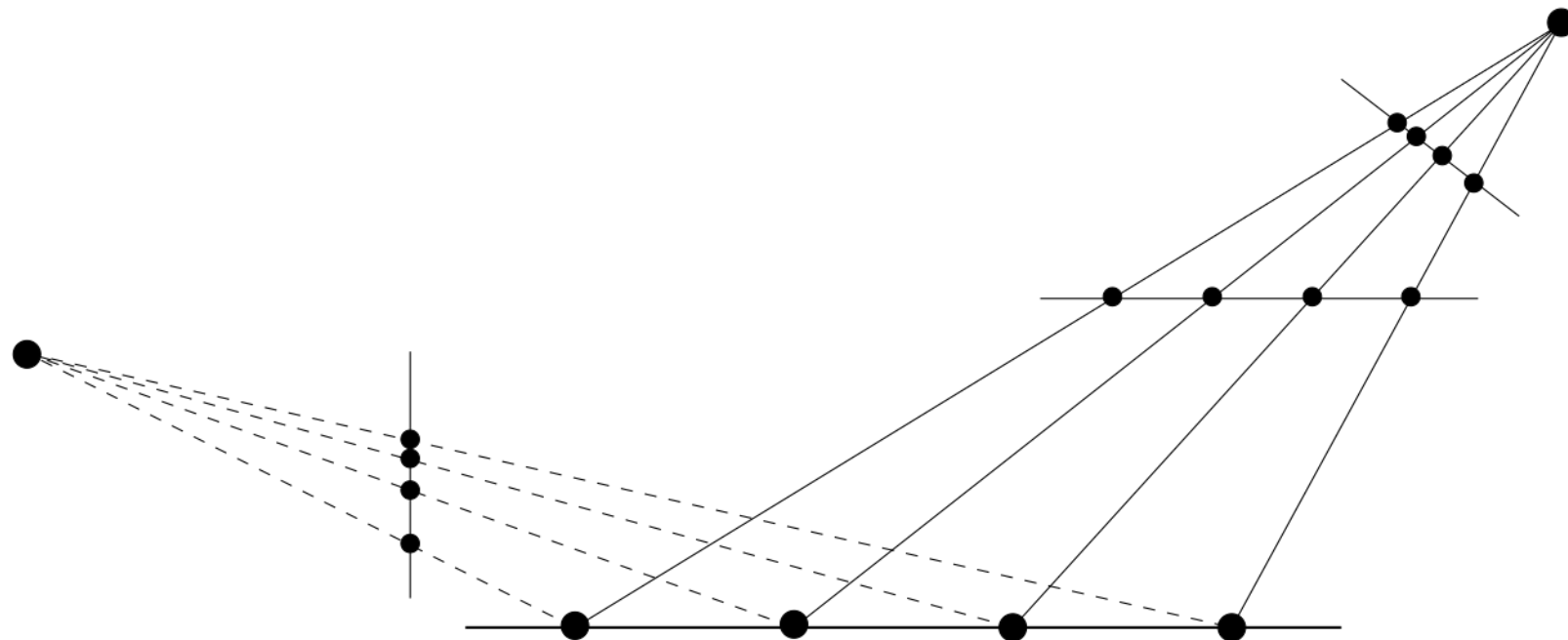


Fig. 2.8. **Projective transformations between lines.** *There are four sets of four collinear points in this figure. Each set is related to the others by a line-to-line projectivity. Since the cross ratio is an invariant under a projectivity, the cross ratio has the same value for all the sets shown.*

Projective geometry of 1D

- The cross ratio

$$\text{Cross}(\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \bar{\mathbf{x}}_3, \bar{\mathbf{x}}_4) = \frac{|\bar{\mathbf{x}}_1 \bar{\mathbf{x}}_2| |\bar{\mathbf{x}}_3 \bar{\mathbf{x}}_4|}{|\bar{\mathbf{x}}_1 \bar{\mathbf{x}}_3| |\bar{\mathbf{x}}_2 \bar{\mathbf{x}}_4|}$$

$$|\bar{\mathbf{x}}_i \bar{\mathbf{x}}_j| = \det \begin{bmatrix} x_{i1} & x_{j1} \\ x_{i2} & x_{j2} \end{bmatrix}$$

Invariant under projective transformations

$$\text{Cross}(\bar{\mathbf{x}}'_1, \bar{\mathbf{x}}'_2, \bar{\mathbf{x}}'_3, \bar{\mathbf{x}}'_4) = \text{Cross}(\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \bar{\mathbf{x}}_3, \bar{\mathbf{x}}_4)$$

$$\bar{\mathbf{x}}' = \mathbf{H}_{2 \times 2} \bar{\mathbf{x}}$$

Projective geometry of 1D

- Concurrent lines

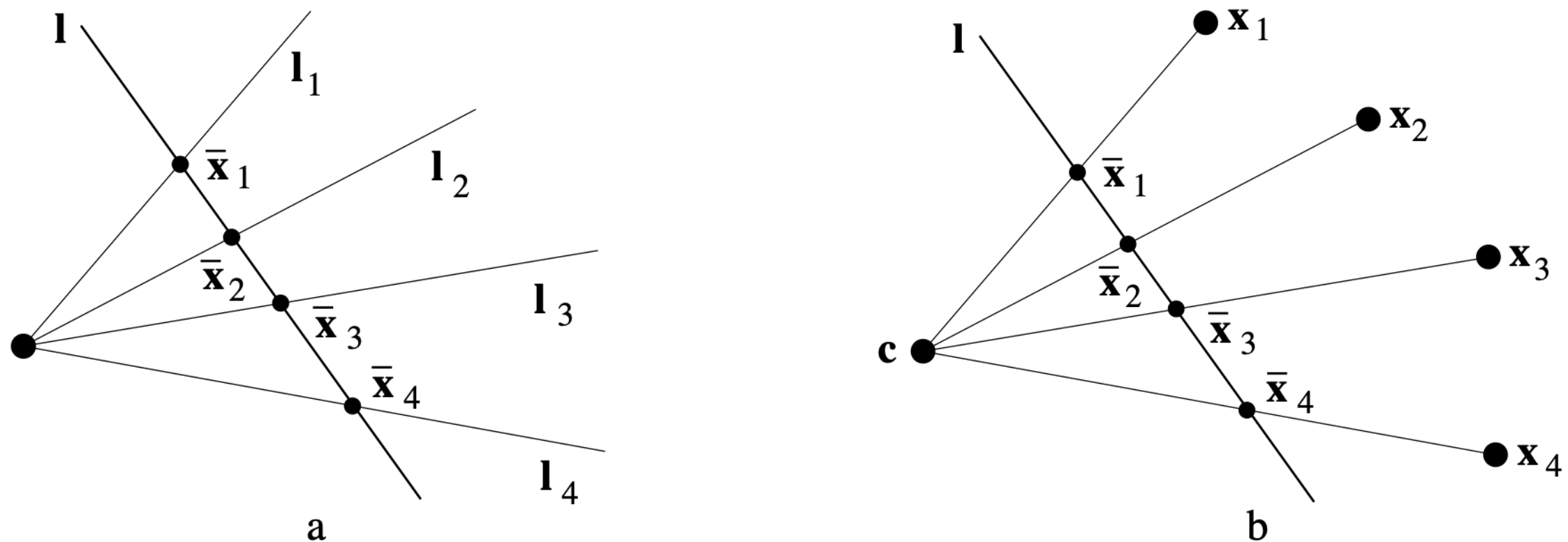


Fig. 2.9. **Concurrent lines.** (a) Four concurrent lines l_i intersect the line l in the four points $\bar{x}_i$. The cross ratio of these lines is an invariant to projective transformations of the plane. Its value is given by the cross ratio of the points, $\text{Cross}(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4)$. (b) Coplanar points x_i are imaged onto a line l (also in the plane) by a projection with centre c . The cross ratio of the image points $\bar{x}_i$ is invariant to the position of the image line l .

Recovering metric and affine properties from images

- Recovery of affine

Remove projective distortion in the perspective image to the extent that affine properties could be measured:
parallelism, ratio of lengths on parallel lines

Identify the line at infinity

- Recovery of metric

Recover the metric properties from a projective distorted image:
angle, ratio of lengths

Identify the circular points

The line at infinity

$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ 0 \end{pmatrix}$$

Affine transformation maps ideal points to ideal points

The line at infinity

$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$

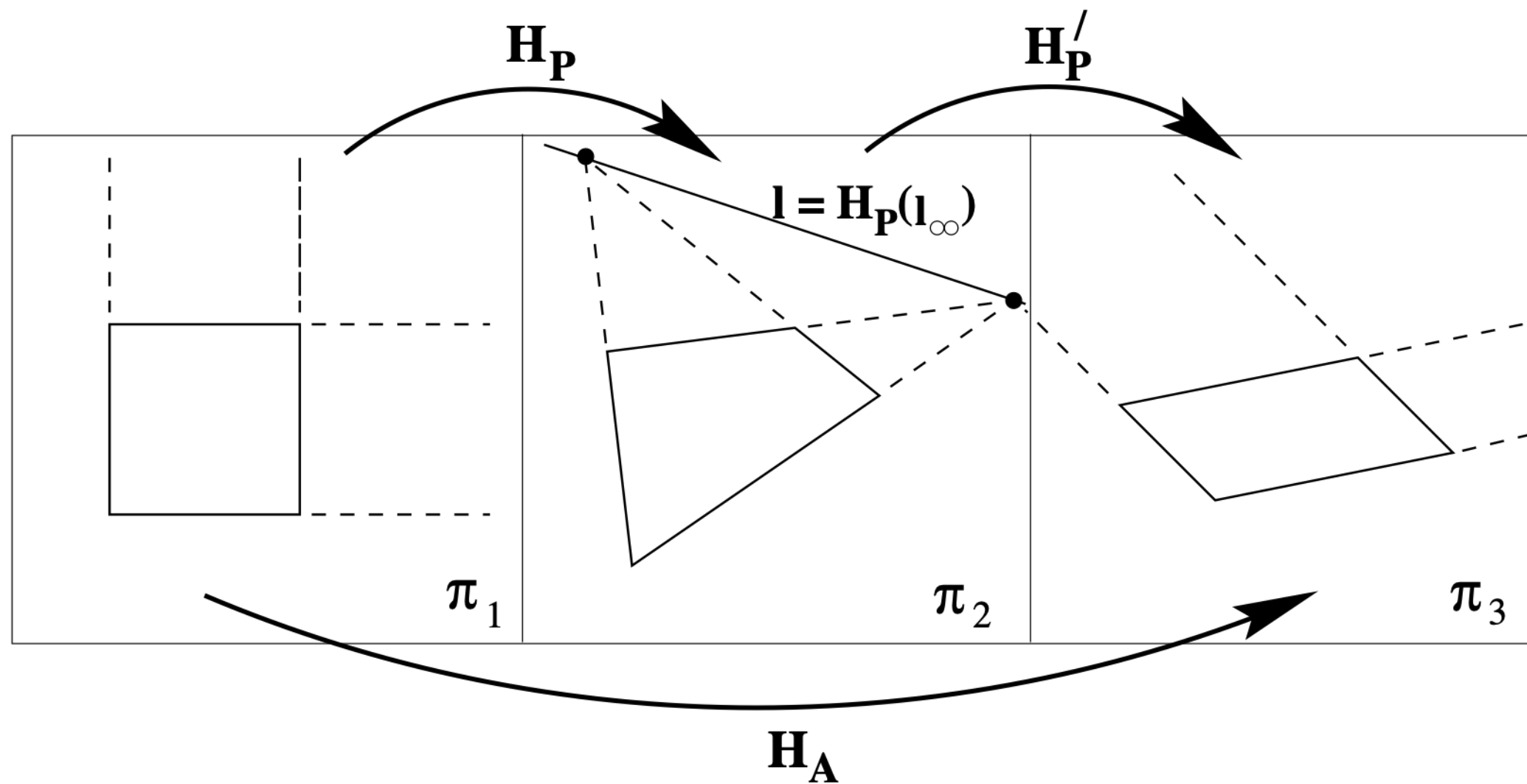
$$\mathbf{l}'_\infty = \mathbf{H}_A^{-\top} \mathbf{l}_\infty = \begin{bmatrix} \mathbf{A}^{-\top} & \mathbf{0} \\ -\mathbf{t}^\top \mathbf{A}^{-\top} & 1 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{l}_\infty$$

Result 2.17. *The line at infinity, $\mathbf{l}_\infty$, is a fixed line under the projective transformation $\mathbf{H}$ if and only if $\mathbf{H}$ is an affinity.*

Not fixed pointwisely unless

$$\mathbf{A}(x_1, x_2)^\top = k(x_1, x_2)^\top$$

Recovery of affine



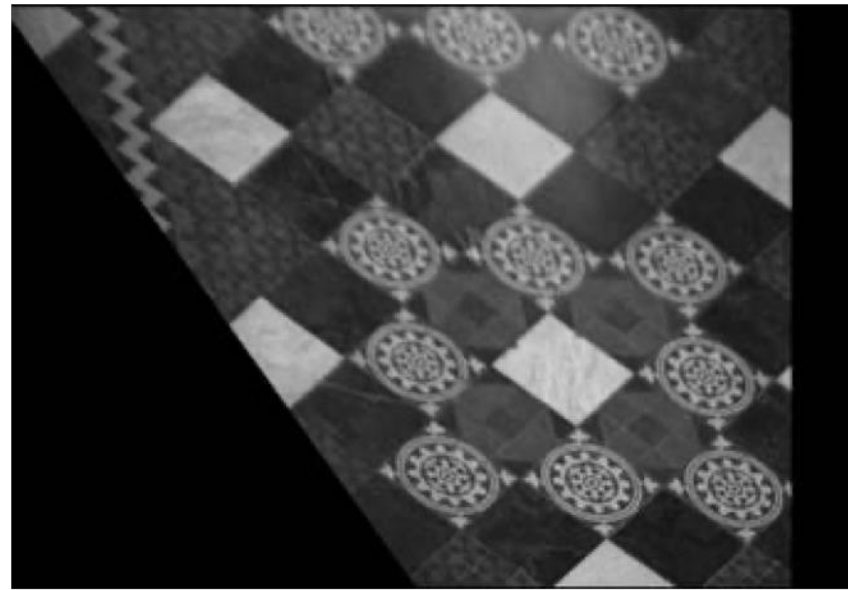
$$H = H_A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$$

$$H^{-T}(l_1, l_2, l_3)^T = (0, 0, 1)^T = l_\infty.$$

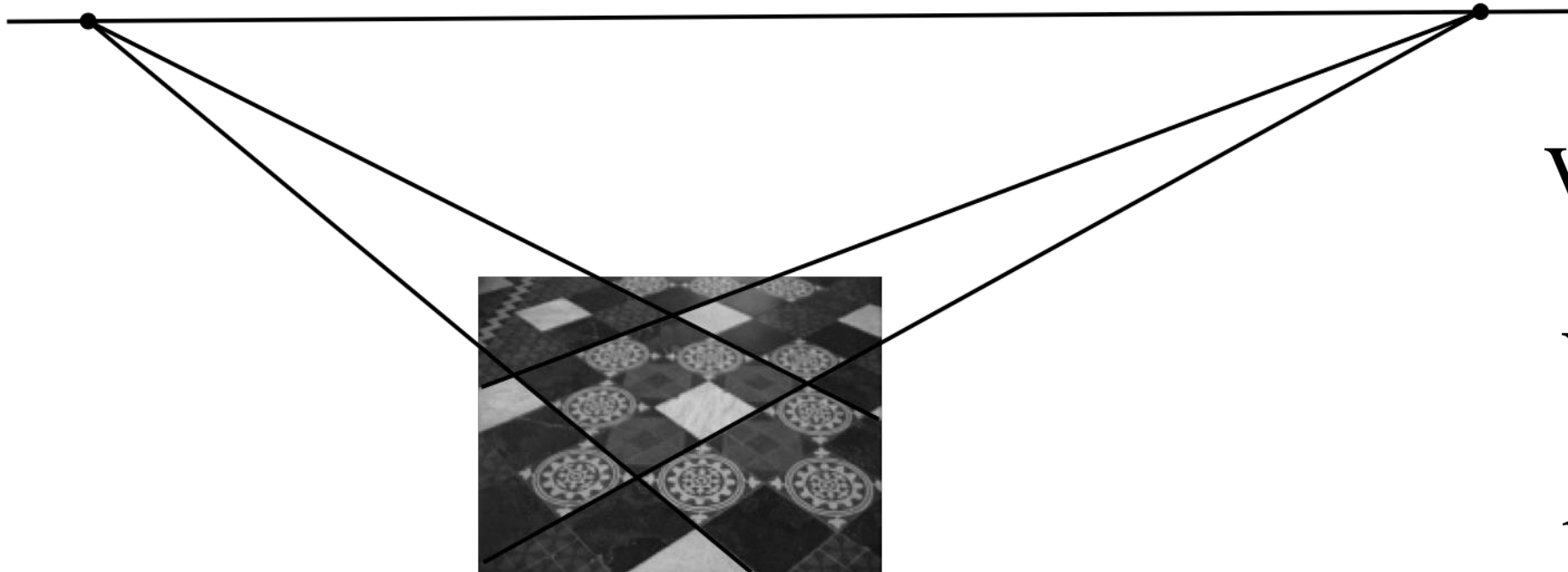
Recovery of affine



a



b



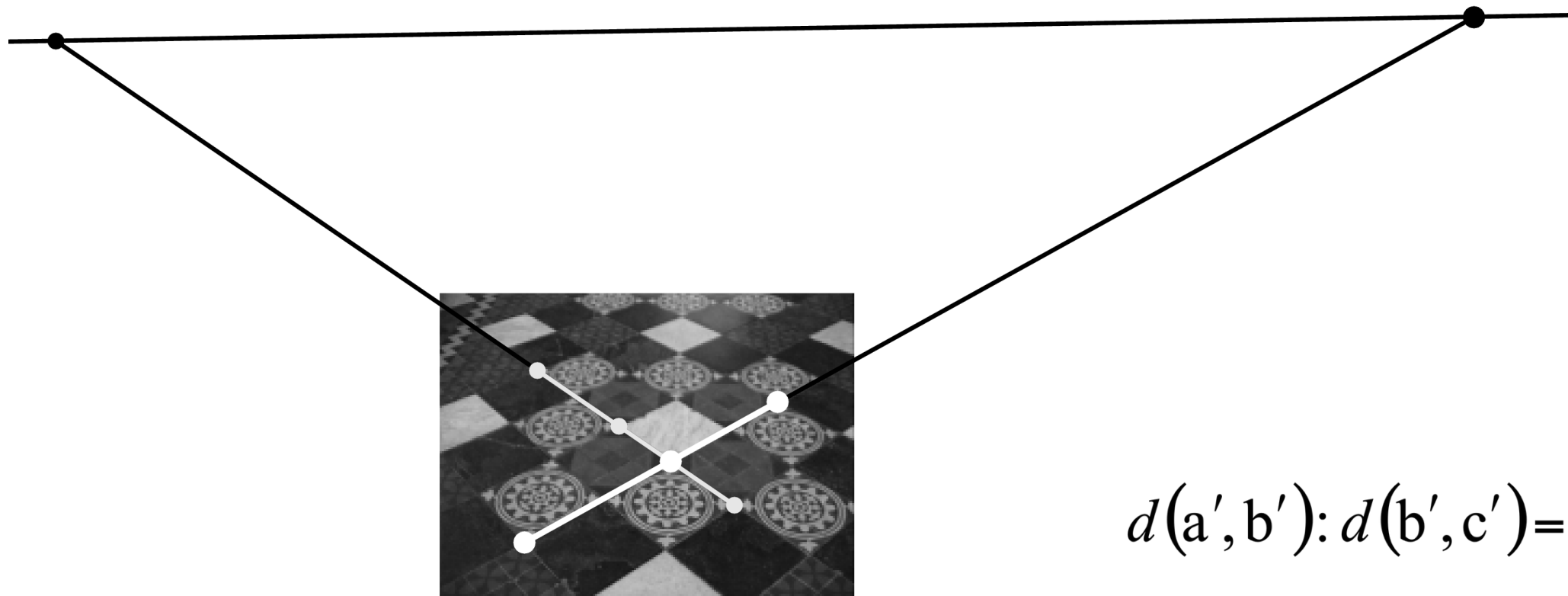
c

$$\mathbf{v}_2 = \mathbf{l}_3 \times \mathbf{l}_4$$

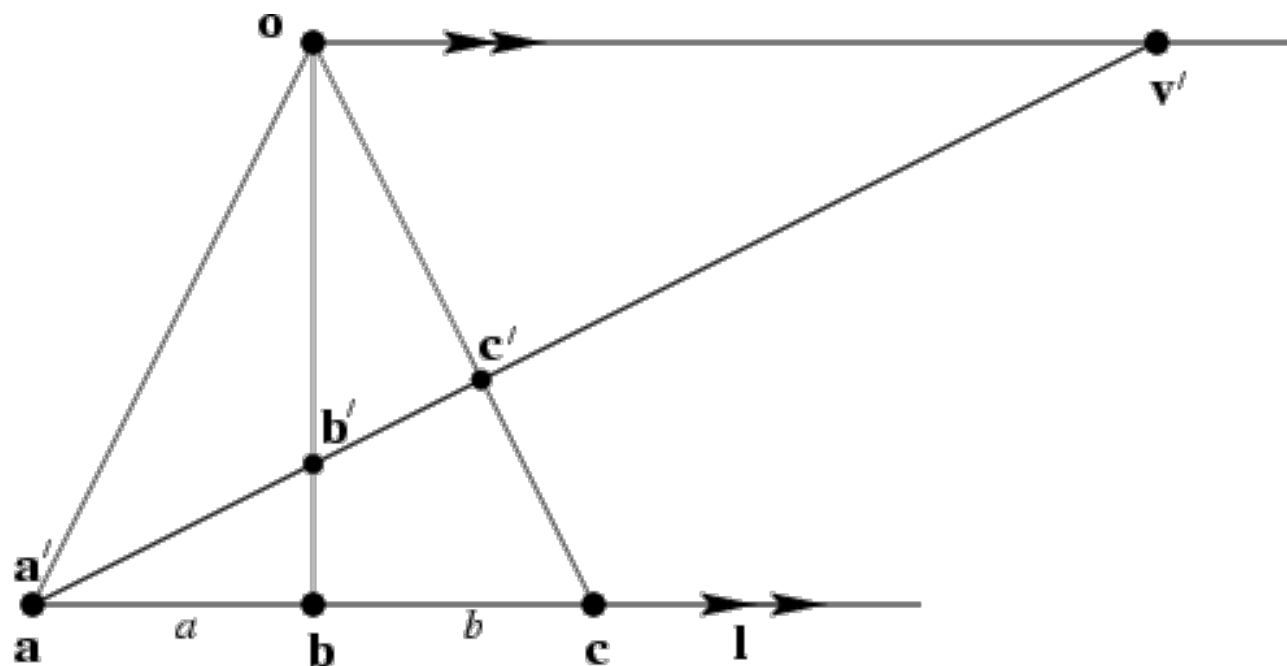
$$\mathbf{v}_1 = \mathbf{l}_1 \times \mathbf{l}_2$$

$$\mathbf{l}_\infty = \mathbf{v}_1 \times \mathbf{v}_2$$

Recovery of affine



$$d(a', b') : d(b', c') = a' : b'$$



$$(0,1)^\top, (a,1)^\top, (a+b,1)^\top$$

$$\begin{array}{c} \downarrow \mathbf{H} \\ \mathbf{a', b', c'} \end{array}$$

$$\mathbf{v}' = \mathbf{H}(1,0)^\top$$

The circular points

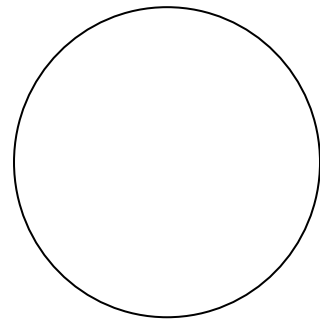
- The absolute points: with canonical coordinates

$$\mathbf{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \qquad \mathbf{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

A pair of complex conjugate ideal points

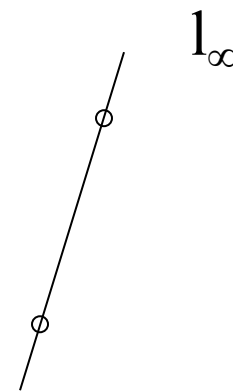
The circular points

Every circle intersects the infinite line at the circular points



$$x_1^2 + x_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

$$x_3 = 0$$



$$x_1^2 + x_2^2 = 0$$

$$I = (1, 0, 0)^T + i(0, 1, 0)^T$$

$$I = (1, i, 0)^T$$

$$J = (1, -i, 0)^T$$

Once the circular points are identified, orthogonality and other metric properties are determined

The circular points

Result 2.21. *The circular points, $\mathbf{I}$, $\mathbf{J}$, are fixed points under the projective transformation $\mathbf{H}$ if and only if $\mathbf{H}$ is a similarity.*

$$\begin{aligned}\mathbf{I}' &= \mathbf{H}_S \mathbf{I} \\ &= \begin{bmatrix} s \cos \theta & -s \sin \theta & t_x \\ s \sin \theta & s \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \\ &= s e^{-i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathbf{I}\end{aligned}$$

The same for $\mathbf{J}$.

The conic dual to the circular points

$$\mathbf{C}_{\infty}^* = \mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T$$

$$\mathbf{C}_{\infty}^* = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \begin{pmatrix} 1 & -i & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} \begin{pmatrix} 1 & i & 0 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A degenerate line conic which consists of the two circular points

4 dof. 5 for a conic - 1 due to degenerate (rank = 2)

The conic dual to the circular points

Result 2.22. *The dual conic C_{∞}^* is fixed under the projective transformation H if and only if H is a similarity.*

Under point transformation $\mathbf{x}' = H_S \mathbf{x}$

$$C_{\infty}^{*'} = H_S C_{\infty}^* H_S^T = C_{\infty}^*$$

The line at infinity is a null vector of the dual conic

$$\mathbf{I}^T \mathbf{l}_{\infty} = \mathbf{J}^T \mathbf{l}_{\infty} = 0$$

$$C_{\infty}^* \mathbf{l}_{\infty} = (\mathbf{I}\mathbf{J}^T + \mathbf{J}\mathbf{I}^T) \mathbf{l}_{\infty} = \mathbf{I}(\mathbf{J}^T \mathbf{l}_{\infty}) + \mathbf{J}(\mathbf{I}^T \mathbf{l}_{\infty}) = \mathbf{0}$$

Angle on projective plane

$$\mathbf{l} = (l_1, l_2, l_3)^\top \quad \mathbf{m} = (\bar{m}_1, m_2, m_3)^\top$$

Euclidean space:

$$\cos \theta = \frac{l_1 m_1 + l_2 m_2}{\sqrt{(l_1^2 + l_2^2)(m_1^2 + m_2^2)}}$$

Projective space:

$$\cos \theta = \frac{\mathbf{l}^\top \mathbf{C}_\infty^* \mathbf{m}}{\sqrt{(\mathbf{l}^\top \mathbf{C}_\infty^* \mathbf{l})(\mathbf{m}^\top \mathbf{C}_\infty^* \mathbf{m})}}$$

It is invariant to projective transformation

Angle on projective plane

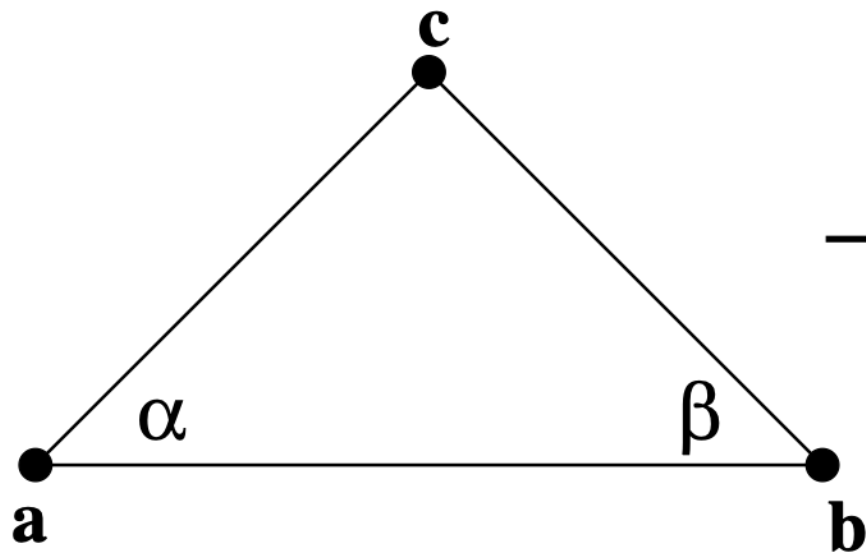
Result 2.23. *Once the conic C_{∞}^* is identified on the projective plane then Euclidean angles may be measured by (2.22).*

$$\cos \theta = \frac{\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m}}{\sqrt{(\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{l})(\mathbf{m}^T \mathbf{C}_{\infty}^* \mathbf{m})}}$$

Result 2.24. *Lines $\mathbf{l}$ and $\mathbf{m}$ are orthogonal if $\mathbf{l}^T \mathbf{C}_{\infty}^* \mathbf{m} = 0$.*

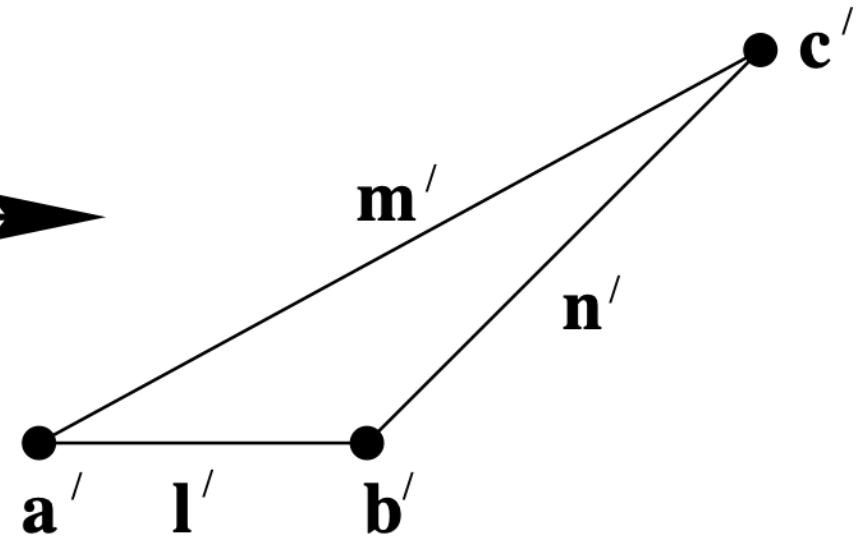
The lines are conjugate with respect to the dual conic.

Length ratio



Euclidean space

$$\frac{d(b, c)}{d(a, c)} = \frac{\sin \alpha}{\sin \beta}$$



Projective space

$$\cos \theta = \frac{l'^T C_{\infty}^* m}{\sqrt{(l'^T C_{\infty}^* l')(m'^T C_{\infty}^* m)}}$$

$$l' = a' \times b'$$

$$m' = c' \times a'$$

$$n' = b' \times c'$$

Recovery of metric properties

Result 2.25. *Once the conic C_∞^* is identified on the projective plane then projective distortion may be rectified up to a similarity.*

$$H = H_P H_A H_S = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

$$\begin{aligned} C_\infty^{*'} &= (H_P H_A H_S) C_\infty^* (H_P H_A H_S)^\top = (H_P H_A) (H_S C_\infty^* H_S^\top) (H_A^\top H_P^\top) \\ &= (H_P H_A) C_\infty^* (H_A^\top H_P^\top) \\ &= \begin{bmatrix} \mathbf{K}\mathbf{K}^\top & \mathbf{K}\mathbf{K}^\top \mathbf{v} \\ \mathbf{v}^\top \mathbf{K}\mathbf{K}^\top & \mathbf{v}^\top \mathbf{K}\mathbf{K}^\top \mathbf{v} \end{bmatrix}. \end{aligned}$$

Rectify the transformation from SVD

$$C_\infty^{*'} = \mathbf{U} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{U}^\top \qquad \mathbf{H} = \mathbf{U}$$

Recovery of metric properties

Example



a



b

$$\mathbf{l}^\top \mathbf{C}_\infty^* \mathbf{m} = 0,$$

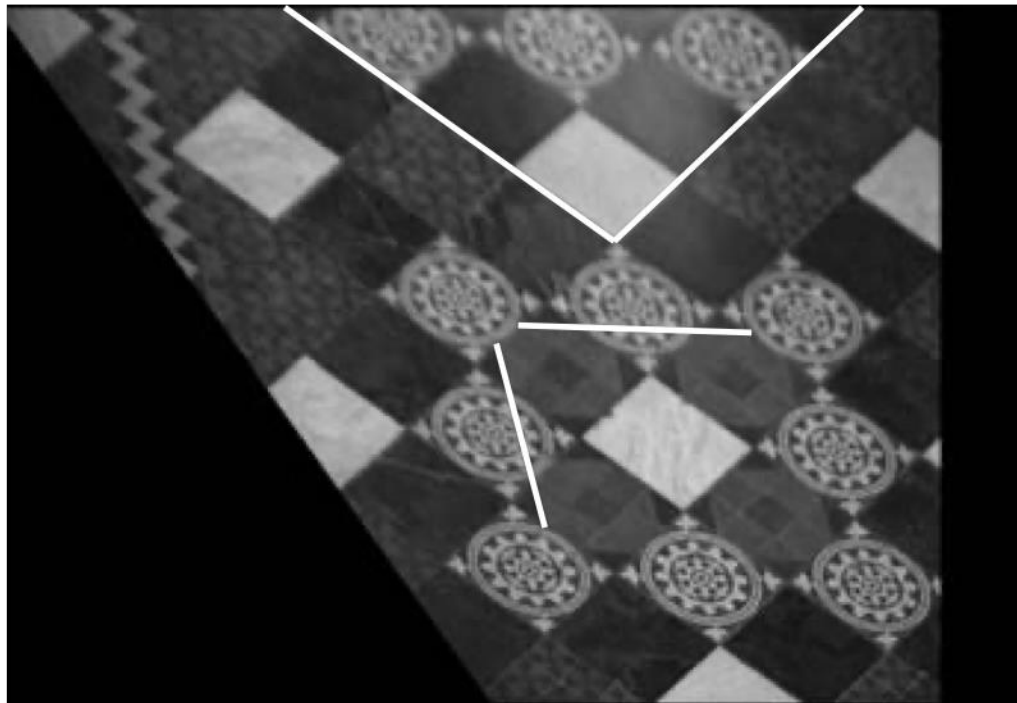
$$(l_1 m_1, (l_1 m_2 + l_2 m_1)/2, l_2 m_2, (l_1 m_3 + l_3 m_1)/2, (l_2 m_3 + l_3 m_2)/2, l_3 m_3) \mathbf{c} = 0$$

$$\mathbf{c} = (a, b, c, d, e, f)^\top$$

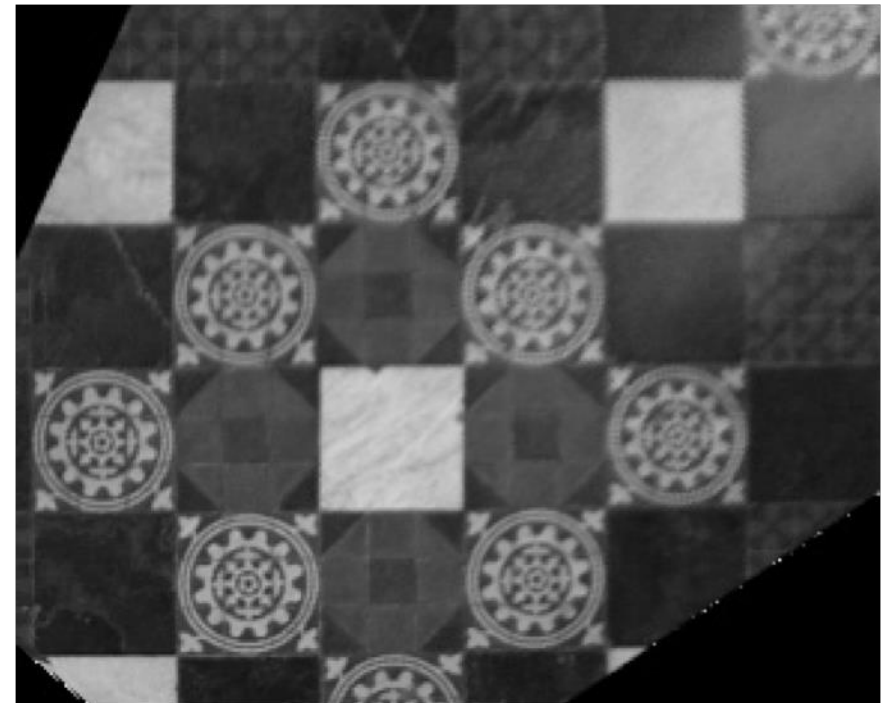
5 pairs of orthogonal lines $\longrightarrow$ 5 constraints on the dual conic

Recovery of metric properties

Stratification



a



b

Step 1: projective to affine, remove affine distortion

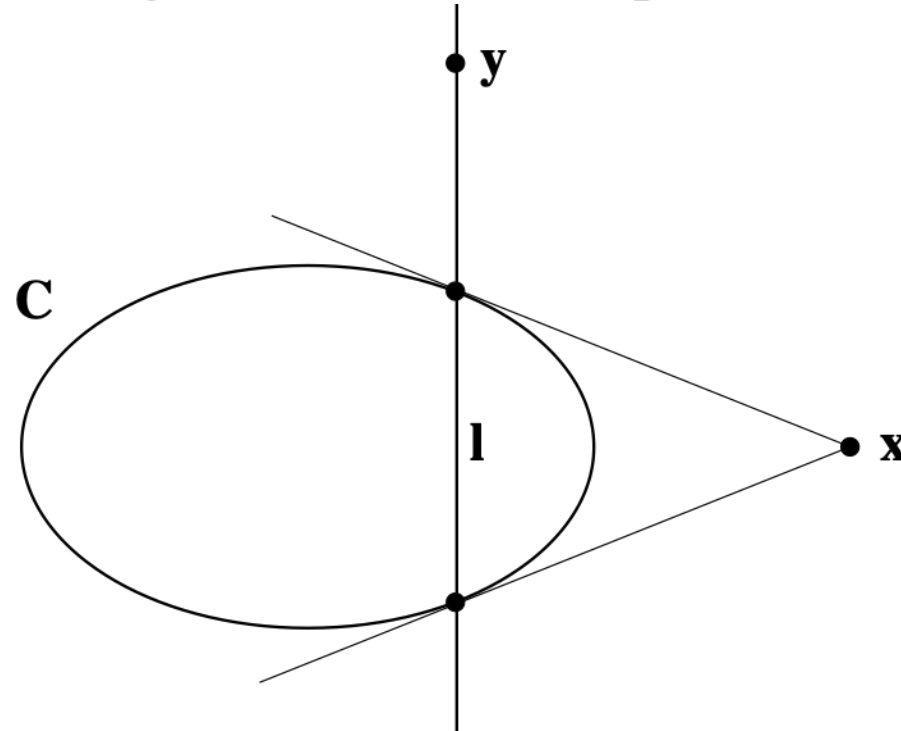
Step 2: affine to Euclidean, remove metric distortion

More properties of conics

Pole-polar relationship

A point x and conic C define a line $l = Cx$. The line l is called the *polar* of x with respect to C , and the point x is the *pole* of l with respect to C .

The polar line $l = Cx$ of the point x with respect to a conic C intersects the conic in two points. The two lines tangent to C at these points intersect at x .

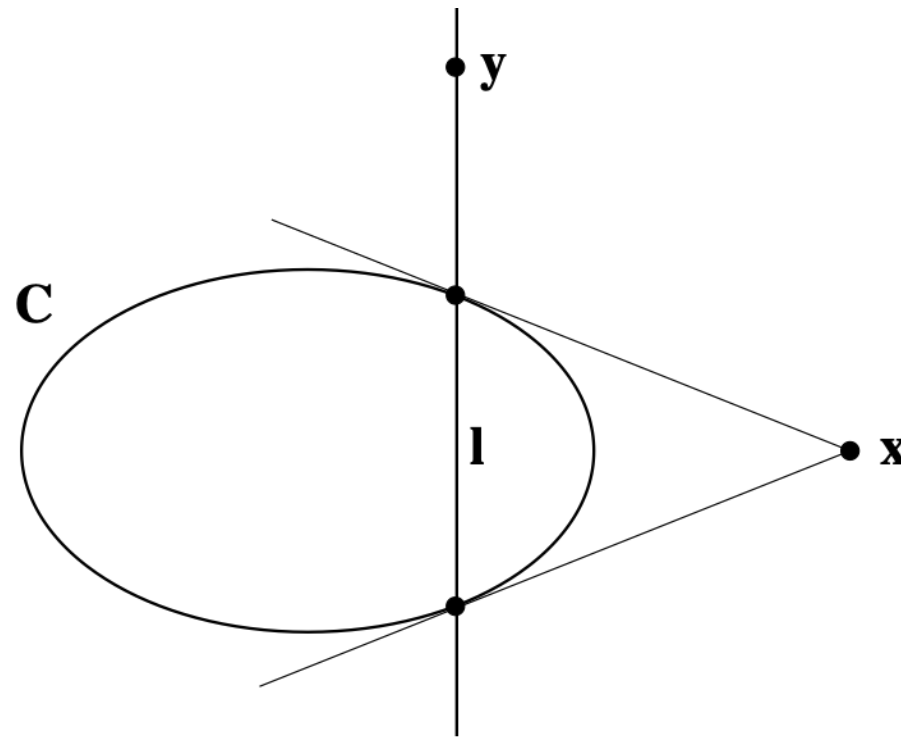


If the point x is on C then the polar is the tangent line to the conic at x .

More properties of conics

Pole-polar relationship

Conjugate points. *If the point y is on the line $l = Cx$ then $y^T l = y^T Cx = 0$. Any two points x, y satisfying $y^T Cx = 0$ are conjugate with respect to the conic C .*



If x is on the polar of y then y is on the polar of x .

$$x^T Cy = y^T Cx$$

Conjugate points lie on each other's polar

Classification of conics

Conic C is a symmetric matrix with real eigen-values

$$C = U^T D U$$

Apply the projective transformation represented by U

$$C' = U^{-T} C U^{-1} = U^{-T} U^T D U U^{-1} = D.$$

Diagonal	Equation	Conic type
$(1, 1, 1)$	$x^2 + y^2 + w^2 = 0$	Improper conic – no real points.
$(1, 1, -1)$	$x^2 + y^2 - w^2 = 0$	Circle
$(1, 1, 0)$	$x^2 + y^2 = 0$	Single real point $(0, 0, 1)^T$
$(1, -1, 0)$	$x^2 - y^2 = 0$	Two lines $x = \pm y$
$(1, 0, 0)$	$x^2 = 0$	Single line $x = 0$ counted twice.

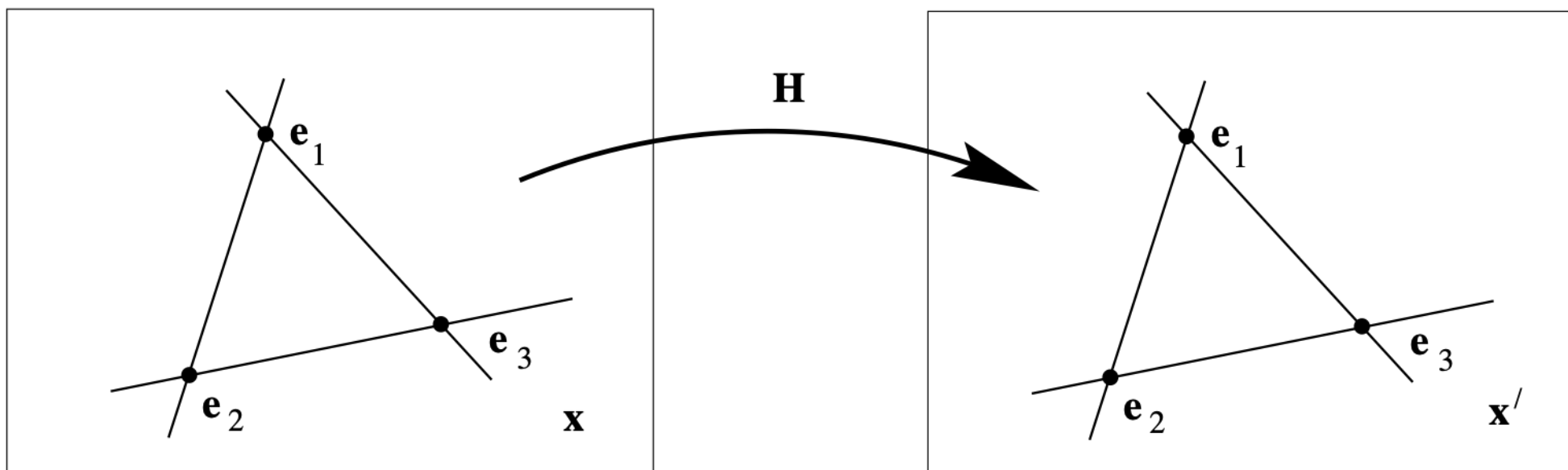
Fixed points and lines

The points and lines may be fixed under a projective transformation
(the circular points, the line at infinity)

$$\mathbf{H}\mathbf{e} = \lambda\mathbf{e} \quad (\text{eigenvectors } \mathbf{H} = \text{fixed points})$$

The eigenvector corresponds to a fixed point of the transformation

$$\mathbf{l}' = \mathbf{H}^{-\top}\mathbf{l} \quad (\text{eigenvectors } \mathbf{H}^{-\top} = \text{fixed lines})$$



Recovering metric and affine properties from images

- Recovery of affine

Remove projective distortion in the perspective image to the extent that affine properties could be measured:
parallelism, ratio of lengths on parallel lines

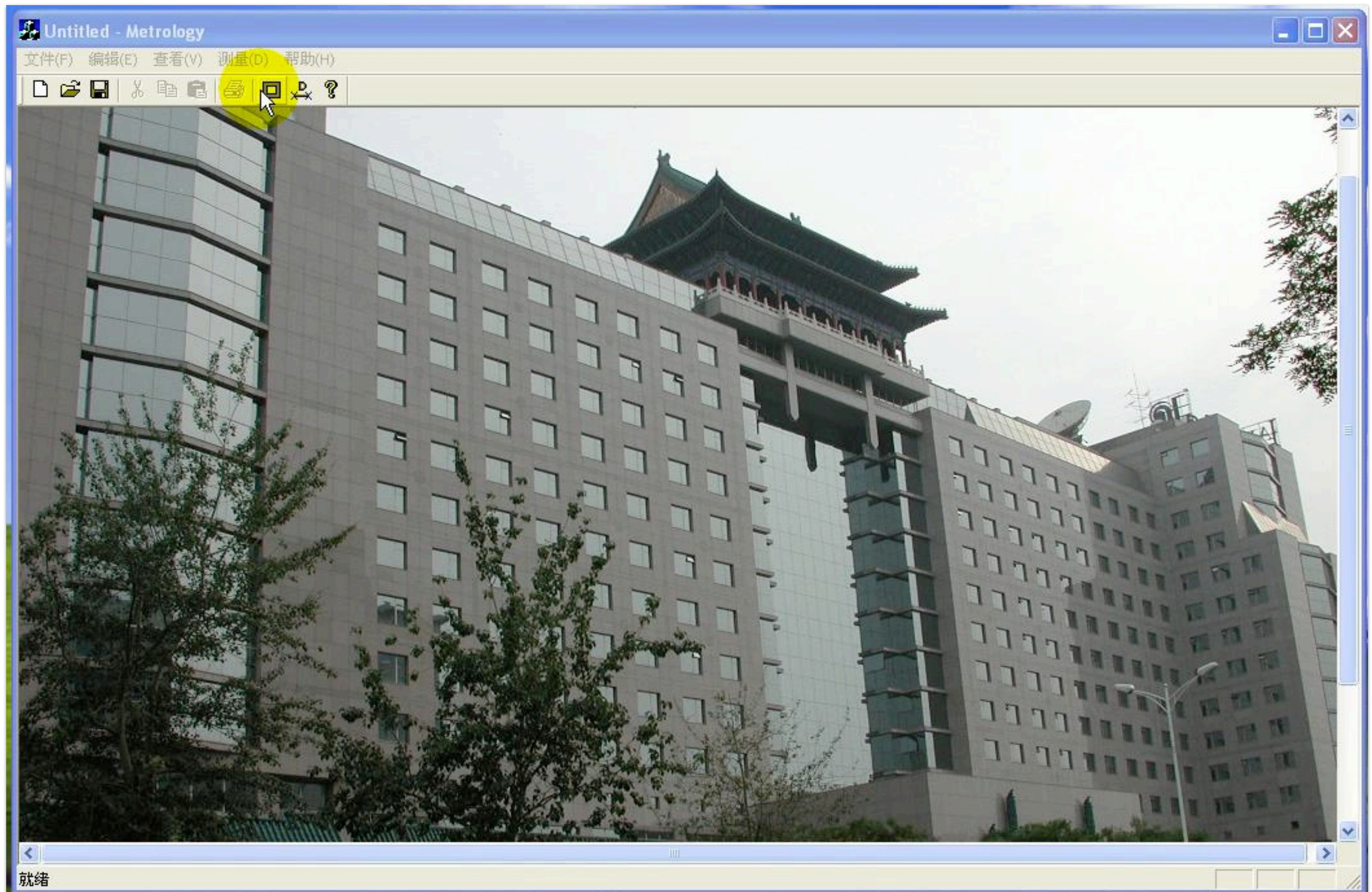
Identify the line at infinity

- Recovery of metric

Recover the metric properties from a projective distorted image:
angle, ratio of lengths

Identify the circular points

Single View Metrology



Single View Metrology



Wang, G., Hu, Z., et al. (2005). [Single view metrology from scene constraints](#). *Image and Vision Computing*, 23(9), 831-840.

Summary

- Homogeneous representation of lines and points
- Ideal points and line at infinity
- Duality principle
- Conics and dual conics
- Projective transformation
- Transformation of lines and conics

Summary

- Projective transformation
- Transformation of lines and conics
- A hierarchy of transformation and invariant properties
 - Projective transformation — 8 dof
 - Affine transformation — 6 dof
 - Similarity transformation — 4 dof
 - Euclidean transformation — 3 dof

Summary

- Recovery of affine properties: line at infinity
- Circular points and the dual conic
- Recovery of metric properties: the circular points
- More properties of conics

Summary

- Some online resources of projective geometry:

Projective geometry - Wikipedia, the free encyclopedia

http://en.wikipedia.org/wiki/Projective_geometry

An Introduction to Projective Geometry (for computer vision)

<http://robotics.stanford.edu/~birch/projective/>

A course on projective geometry

https://people.maths.ox.ac.uk/hitchin/hitchinnotes/Projective_geometry/

Projective Geometry: From Foundations to Applications

<http://www.maths.ed.ac.uk/~aar/papers/beutel.pdf>

Projective Geometry for Machine Vision

<http://www.cs.cmu.edu/~ph/869/papers/zisser-mundy.pdf>

THANK YOU